Software Performance Engineering

Lecture 7
C to Assembly

Xuhao Chen

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Context for This Lecture

Lectures 4 & 5: Computer Arch and Vectorization

- x86-64 instructions, registers, addressing modes, condition codes, SSE, AVX, etc

C code fib.c

```c
int64_t fib(int64_t n) {
  if (n < 2) return n;
  return fib(n-1) + fib(n-2);
}
```

```
$ clang -O1 fib.c -S
```

Assembly code fib.s

```
            .globl     _fib
            .p2align   4, 0x90
_fib:       ## @fib
            pushq      %rbp
            movq       %rsp, %rbp
            pushq      %r14
            pushq      %rbx
            movq       %rdi, %rbx
            cmpq       $2, %rdi
            jl         LBB0_2
            leaq       -1(%rbx), %rdi
            callq      _fib
            movq       %rax, %r14
            addq       $-2, %rbx
            movq       %rbx, %rdi
            callq      _fib
            movq       %rax, %rbx
            addq       %r14, %rbx
LBB0_2:
            movq       %rbx, %rax
            popq       %rbx
            popq       %r14
            popq       %rbp
            retq
```

Mapping C Code to Assembly

This lecture:

- How C code gets translated into assembly
- Unclear how C code relates to assembly!

C code `fib.c`

```c
int64_t fib(int64_t n) {
  if (n < 2) return n;
  return fib(n-1) + fib(n-2);
}
```

```
$ clang -O1 fib.c -S
```

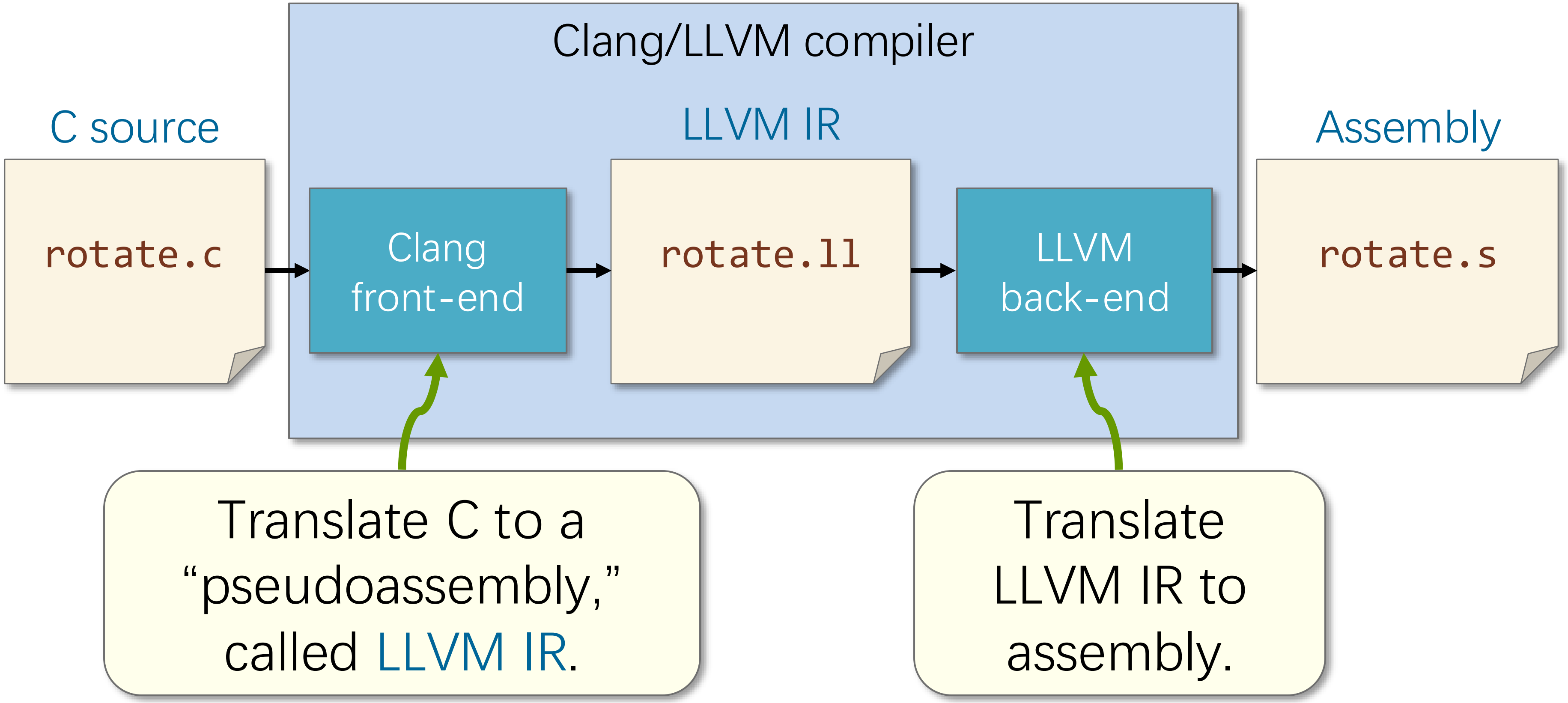
?

Assembly code `fib.s`

```
        .globl      _fib
        .p2align    4, 0x90
_fib:   ## @fib
        pushq       %rbp
        movq        %rsp, %rbp
        pushq       %r14
        pushq       %rbx
        movq        %rdi, %rbx
        cmpq        $2, %rdi
        jl          LBB0_2
        leaq        -1(%rbx), %rdi
        callq       _fib
        movq        %rax, %r14
        addq        $-2, %rbx
        movq        %rbx, %rdi
        callq       _fib
        movq        %rax, %rbx
        addq        %r14, %rbx
LBB0_2:
        movq        %rbx, %rax
        popq        %rbx
        popq        %r14
        popq        %rbp
        retq
```

Clang/LLVM Compiler Pipeline

To understand this mapping, let us see how the compiler reasons about it.



Software Performance Engineering

LECTURE 7

C to LLVM IR to Assembly

Running Example: fib.c

The C function fib computes the nth Fibonacci number F(n) recursively using the formula:

$$F(n) = \begin{cases} n & \text{if } n \in \{0,1\} \\ F(n-1) + F(n-2) & \text{otherwise.} \end{cases}$$

C code fib.c

```
int64_t fib(int64_t n) {
  if (n < 2)
    return n;
  return fib(n-1) + fib(n-2);
}
```

Outline

- Running Example: Fib
- C to LLVM IR
  - Basic Organization of LLVM IR
  - C Conditionals in LLVM IR
  - Static Single Assignment (SSA) Form
- LLVM IR to Assembly
  - Linux x86-64 Calling Convention
- Optional Slides

Basic Organization of LLVM IR

From `fib.c` to `fib.ll`

C code `fib.c`

```c
int64_t fib(int64_t n)
{
  if (n < 2)
    return n;
  return
    fib(n-1)
    +
    fib(n-2);
}
```

LLVM IR code `fib.ll`

```llvm
define i64 @fib(i64 %0) {
  %2 = icmp slt i64 %0, 2
  br i1 %2, label %9, label %3

3:                                ; preds = %1
  %4 = add nsw i64 %0, -1
  %5 = call i64 @fib(i64 %4)
  %6 = add nsw i64 %0, -2
  %7 = call i64 @fib(i64 %6)
  %8 = add nsw i64 %7, %5
  br label %9

9:                                ; preds = %1, %3
  %10 = phi i64 [ %8, %3 ], [ %0, %1 ]
  ret i64 %10
}
```

Components of LLVM IR

LLVM IR code `fib.ll`

Function

LLVM IR registers

Instructions

Data types

```
define i64 @fib(i64 %0) {
  %2 = icmp slt i64 %0, 2
  br i1 %2, label %9, label %3

3:                                   ; preds = %1
  %4 = add nsw i64 %0, -1
  %5 = call i64 @fib(i64 %4)
  %6 = add nsw i64 %0, -2
  %7 = call i64 @fib(i64 %6)
  %8 = add nsw i64 %7, %5
  br label %9

9:                                   ; preds = %1, %3
  %10 = phi i64 [ %8, %3 ], [ %0, %1 ]
  ret i64 %10
}
```

Comparing LLVM IR and Assembly

- LLVM IR is similar to assembly
  - LLVM IR uses a simple instruction format, i.e., ⟨destination operand⟩ = ⟨opcode⟩ ⟨source operands⟩
  - LLVM IR is similar in structure to assembly
  - Control flow is implemented using conditional and unconditional branches

- LLVM IR is simpler than assembly.
  - C-like functions
  - Smaller instruction set
  - Infinite registers, i.e. LLVM IR registers are similar to local variables in C
  - No implicit FLAGS register or condition codes
  - No explicit stack pointer or frame pointer

LLVM IR Functions

Functions in LLVM IR resemble functions in C.

C code `fib.c`

```c
int64_t fib(int64_t n) {
  …
  return n;
```

LLVM IR function parameters map directly to their C counterparts.

Function declarations and definitions are C-like.

LLVM IR `fib.ll`

```llvm
define i64 @fib(i64 %0) {
  …
  ret i64 %10
}
```

A `ret` instruction terminates the function, just like a return statement in C.

Straight-Line C Code in LLVM IR

Straight-line C code — code containing no conditionals or loops — becomes a sequence of LLVM IR instructions.

- Intermediate results are stored in registers
- Arguments are evaluated before the C operation

C code

```
fib(n-1) + fib(n-2)
```

LLVM IR (register %0 holds the value of n)

```
%4 = add nsw i64 %0, -1
%5 = call i64 @fib(i64 %4)
%6 = add nsw i64 %0, -2
%7 = call i64 @fib(i64 %6)
%8 = add nsw i64 %7, %5
```

Static Single Assignment

LLVM IR maintains static single assignment (SSA) form:
a register is defined by at most **one** instruction in a function

C code

```
fib(n-1) + fib(n-2)
```

LLVM IR (register `%0` holds the value of `n`)

```
%4 = add nsw i64 %0, -1
%5 = call i64 @fib(i64 %4)
%6 = add nsw i64 %0, -2
%7 = call i64 @fib(i64 %6)
%8 = add nsw i64 %7, %5
```

Why? SSA form makes it easier for the compiler to reason about the program's behavior

We'll look at SSA form more in a few slides.

Basic Blocks

The body of a function definition is partitioned into **basic blocks**:

sequences of instructions where control only enters through the first instruction and only exits from the last

C code `fib.c`

```c
int64_t fib(int64_t n)
{
  if (n < 2)
    return n;
  return
    fib(n-1)+fib(n-2);
}
```

LLVM IR `fib.ll`

```llvm
define i64 @fib(i64 %0) {
  %2 = icmp slt i64 %0, 2
  br i1 %2, label %9, label %3

3:                                   ; preds = %1
  %4 = add nsw i64 %0, -1
  %5 = call i64 @fib(i64 %4)
  %6 = add nsw i64 %0, -2
  %7 = call i64 @fib(i64 %6)
  %8 = add nsw i64 %7, %5
  br label %9

9:                                   ; preds = %1, %3
  %10 = phi i64 [ %8, %3 ], [ %0, %1 ]
  ret i64 %10
}
```

Basic-Block Terminators

A basic block is terminated by a control-flow instruction that describes how control exits that block.

LLVM IR `fib.ll`

```
define i64 @fib(i64 %0) {
  %2 = icmp slt i64 %0, 2
  br i1 %2, label %9, label %3

3:                            ; preds = %1
  %4 = add nsw i64 %0, -1
  %5 = call i64 @fib(i64 %4)
  %6 = add nsw i64 %0, -2
  %7 = call i64 @fib(i64 %6)
  %8 = add nsw i64 %7, %5
  br label %9

9:                            ; preds = %1, %3
  %10 = phi i64 [ %8, %3 ], [ %0, %1 ]
  ret i64 %10
}
```

Control-Flow Graphs (CFGs)

Control-flow instructions (e.g., `br` instructions) induce control-flow edges between basic blocks, creating a control-flow graph (CFG).

Control-flow graph for `@fib`

Block *1*

```
%2 = icmp slt i64 %0, 2
br i1 %2, label %9, label %3
```

True branch

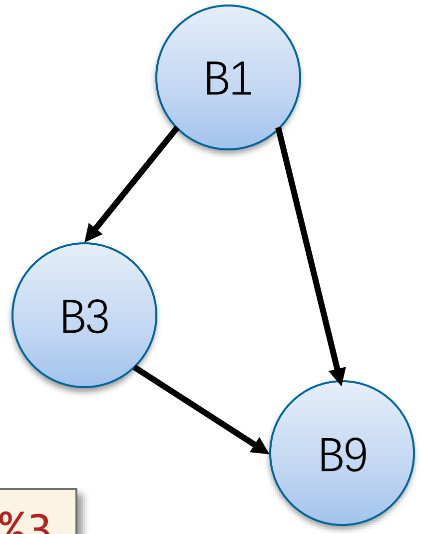
False branch

Block *3*

```
3:                              ; preds = %1
  %4 = add nsw i64 %0, -1
  %5 = call i64 @fib(i64 %4)
  %6 = add nsw i64 %0, -2
  %7 = call i64 @fib(i64 %6)
  %8 = add nsw i64 %7, %5
  br label %9
```

Block *9*

```
9:                              ; preds = %1, %3
  %10 = phi i64 [ %8, %3 ], [ %0, %1 ]
  ret i64 %10
```



C Conditionals in LLVM IR

C Conditionals

A conditional in C is translated into a conditional branch instruction, **br**, in LLVM IR.

C code **fib.c**

```
int64_t fib(int64_t n)
{
  if (n < 2)
    …
  return …
}
```

LLVM IR **fib.ll**

```
define i64 @fib(i64 %0) {
  %2 = icmp slt i64 %0, 2
  br i1 %2, label %9, label %3

3:                        ; preds = %1
  …

9:                        %1, %3
  …
  ret i64 %10
}
```

The `slt` flag means "signed less than."

The comparison in C becomes an `icmp` instruction, with a flag that describes the comparison.

Arguments of a Conditional Branch

Conditional branch in LLVM IR takes as operands a 1-bit integer and two basic-blocks, thereby producing **two** control-flow edges

Predicate

Destination block if the predicate is true.

Destination block if the predicate is false.

LLVM IR `fib.ll`

```
define i64 @fib(i64 %0) {
  %2 = icmp slt i64 %0, 2
  br i1 %2, label %9, label %3

3:                        ; preds = %1
  …

9:                        ; preds = %1, %3
  …
  ret i64 %10
}
```

Unconditional Branches

If a `br` instruction has just one operand, it is an unconditional branch

LLVM IR

```
3:                      ; preds = %1
  …
  %8 = add nsw i64 %7, %5
  br label %9
```

Unconditional branch to basic block 9.

An unconditional branch terminates its basic block and produces one control-flow edge.

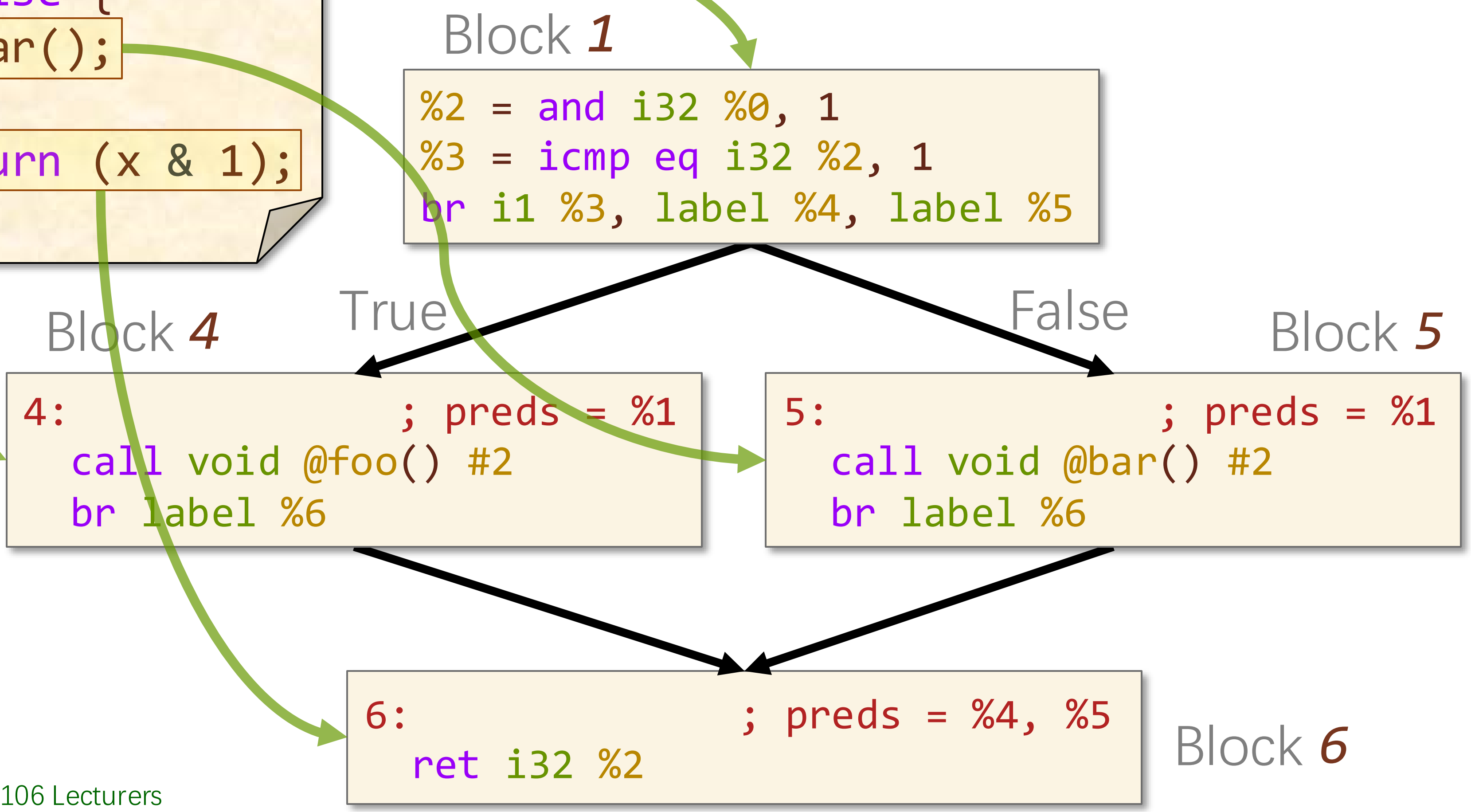
Aside: C Conditionals in General

C code

```c
int baz(int x) {
  if (x & 1) {
    foo();
  } else {
    bar();
  }
  return (x & 1);
}
```

In general, a C conditional creates a diamond pattern in the CFG.

Control-flow graph



STATIC SINGLE ASSIGNMENT FORM

SSA Problem

PROBLEM: How does LLVM represent a variable whose value depends on control flow?

C code

```
int64_t fib(int64_t n) {
  int64_t result;
  if (n < 2)
    result = n;
  else
    result =
        fib(n-1)+fib(n-2);
  return result;
}
```

NOTE: This code generates the same LLVM IR as `fib.c`.

How does LLVM IR express this value, which depends on the conditional?

Control-Flow-Dependent Values

ANSWER: The phi instruction specifies the value of a register depending on control flow.

C code

```c
int64_t fib(int64_t n) {
  int64_t result;
  if (n < 2)
    result = n;
  else
    result =
        fib(n-1)+fib(n-2);
  return result;
}
```

Simplified CFG

Block 1

```
%2 = icmp slt i64 %0, 2
br i1 %2, label %9, label %3
```

Block 3

```
3:                ; preds = %1
  …
  %8 = add nsw i64 %7, %5
  br label %9
```

Block 9

```
9:                          ; preds = %1, %3
  %10 = phi i64 [ %8, %3 ], [ %0, %1 ]
  ret i64 %10
```

The Phi Instruction

The `phi` instruction specifies, for each predecessor *P* of a basic block *B*, the value of the destination register if control enters *B* via *P*.

Block *9*

```
9:                                        ; preds = %1, %3
  %10 = phi i64 [ %8, %3 ], [ %0, %1 ]
  ret i64 %10
```

Adopt the value `%8` if control comes from block 3.

Adopt the value `%0` if control comes from block 1.

Note: `phi` does not correspond to a real instruction in assembly

Summary: C to LLVM IR

- To transform C into LLVM IR:
  - Each function is broken down into basic blocks
  - Control-flow are implemented using branches between basic blocks
  - Other statements and expressions are implemented using primitive operations that may store values in LLVM registers

- Feel free to explore LLVM IR yourself, e.g., with Compiler Explorer

LLVM IR to Assembly

Comparing LLVM IR and Assembly

LLVM IR is structurally similar to assembly.

LLVM IR code `fib.ll`

```
define i64 @fib(i64 %0) {
  %2 = icmp slt i64 %0, 2
  br i1 %2, label %9, label %3

3:                                    ; preds = %1
  %4 = add nsw i64 %0, -1
  %5 = call i64 @fib(i64 %4)
  %6 = add nsw i64 %0, -2
  %7 = call i64 @fib(i64 %6)
  %8 = add nsw i64 %7, %5
  br label %9

9:                                    ; preds = %1, %3
  %10 = phi i64 [ %8, %3 ], [ %0, %1 ]
  ret i64 %10
}
```

Assembly code `fib.s`

```
        .globl      _fib
        .p2align    4, 0x90
_fib:           ## @fib
        pushq       %rbp
        movq        %rsp, %rbp
        pushq       %r14
        pushq       %rbx
        movq        %rdi, %rbx
        cmpq        $2, %rdi
        jl          LBB0_2
        leaq        -1(%rbx), %rdi
        callq       _fib
        movq        %rax, %r14
        addq        $-2, %rbx
        movq        %rbx, %rdi
        callq       _fib
        movq        %rax, %rbx
        addq        %r14, %rbx
LBB0_2:
        movq        %rbx, %rax
        popq        %rbx
        popq        %r14
        popq        %rbp
        retq
```

Translating LLVM IR to Assembly

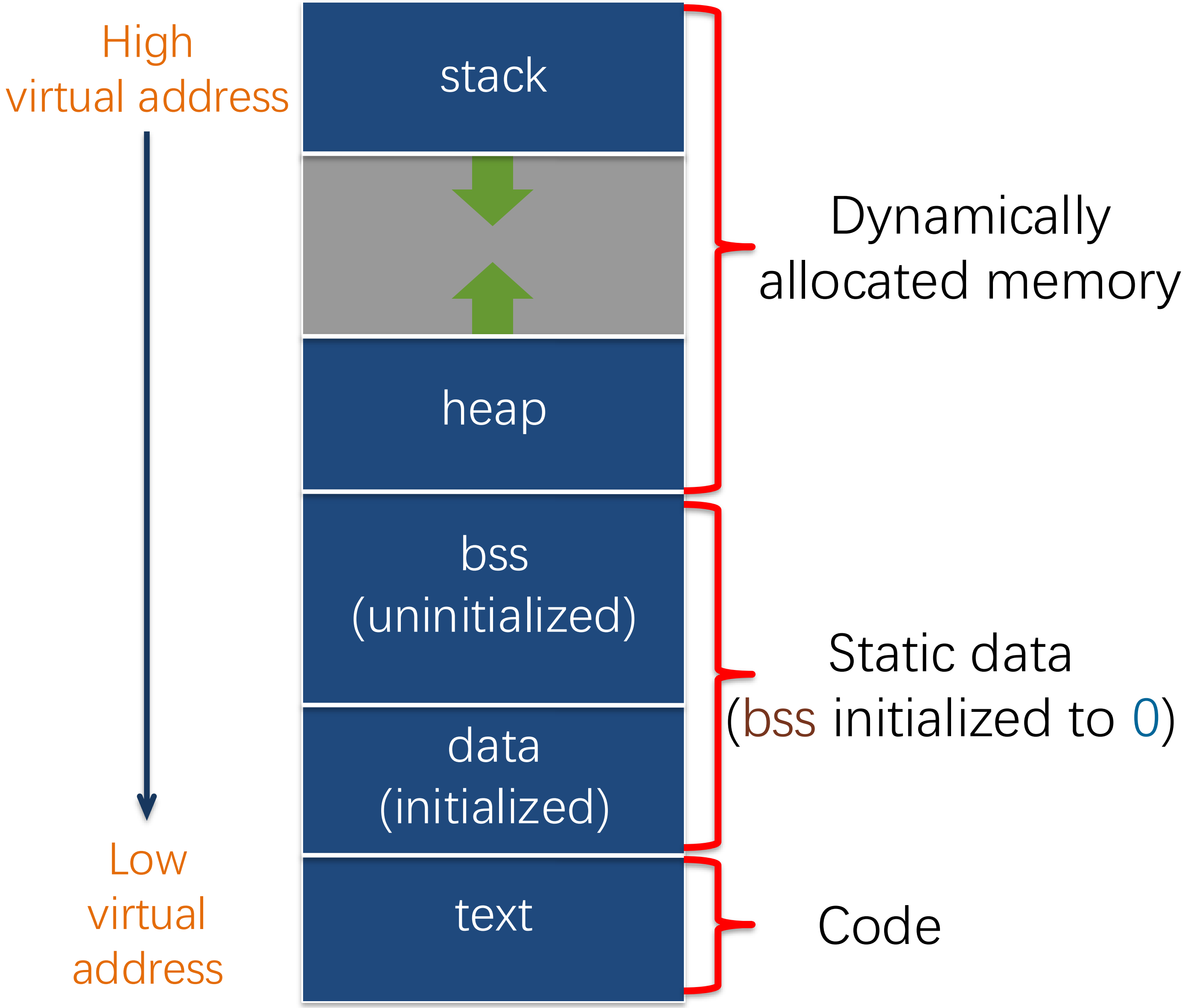
- The compiler does **three tasks** to translate LLVM IR into assembly
  - Select assembly instructions to implement LLVM IR instructions
  - Allocate assembly registers to hold values in LLVM IR registers
  - Coordinate function calls.

Our main focus

The Linux x86-64 Calling Convention

Layout of a Program in Memory

When a program executes, virtual memory is organized into segments.

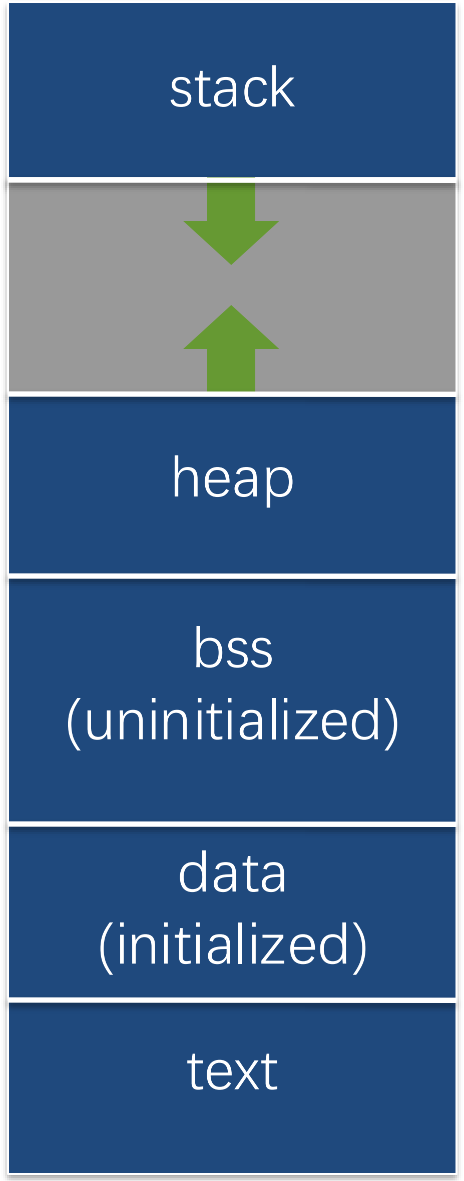


The Stack Segment

The stack segment manages function calls

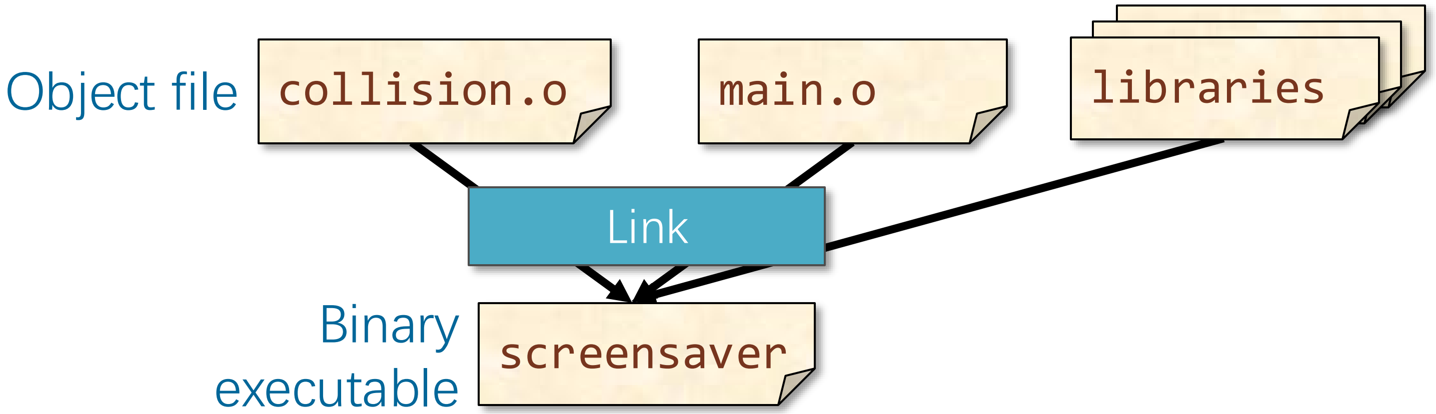
What data is stored on the stack?

- Return addresses of function calls.
- Function arguments and local variables that don't fit in registers.
- Register state, so different functions can use the same registers.



Coordinating Function Calls

PROBLEM: How do functions in different object files coordinate their use of the stack and of the registers?



ANSWER: Functions abide by a calling convention.

Organizing the Stack

- Linux x86-64 calling convention organizes the stack into frames,
- Each function instantiation gets its own frame
  - base (or frame) pointer, `%rbp`, points to the **top** of the current stack frame
  - stack pointer, `%rsp`, points to the **bottom** of the current stack frame

- Stack segment grows **down**, towards lower virtual mem. addresses

Managing Return Addresses

- The `call` and `ret` instructions manage return addresses using the stack and the instruction pointer, `%rip`.
  - A `call` instruction pushes `%rip` onto the stack and jumps to the specified function
  - To return to the caller, a `ret` instruction pops `%rip` from the stack

Maintaining Registers Across Calls

- PROBLEM: Who's responsible for preserving the **register state** across a function call and return?
- Function A calls function B
  - The caller might waste work saving register state that the callee doesn't use
  - The callee might waste work saving register state that the caller wasn't using

- ANSWER: The Linux x86-64 calling convention does a bit of both
  - Callee-saved registers: `%rbx`, `%rbp`, `%r12`-`%r15`.
  - All other registers are caller-saved.

C Linkage for x86-64 GPR's

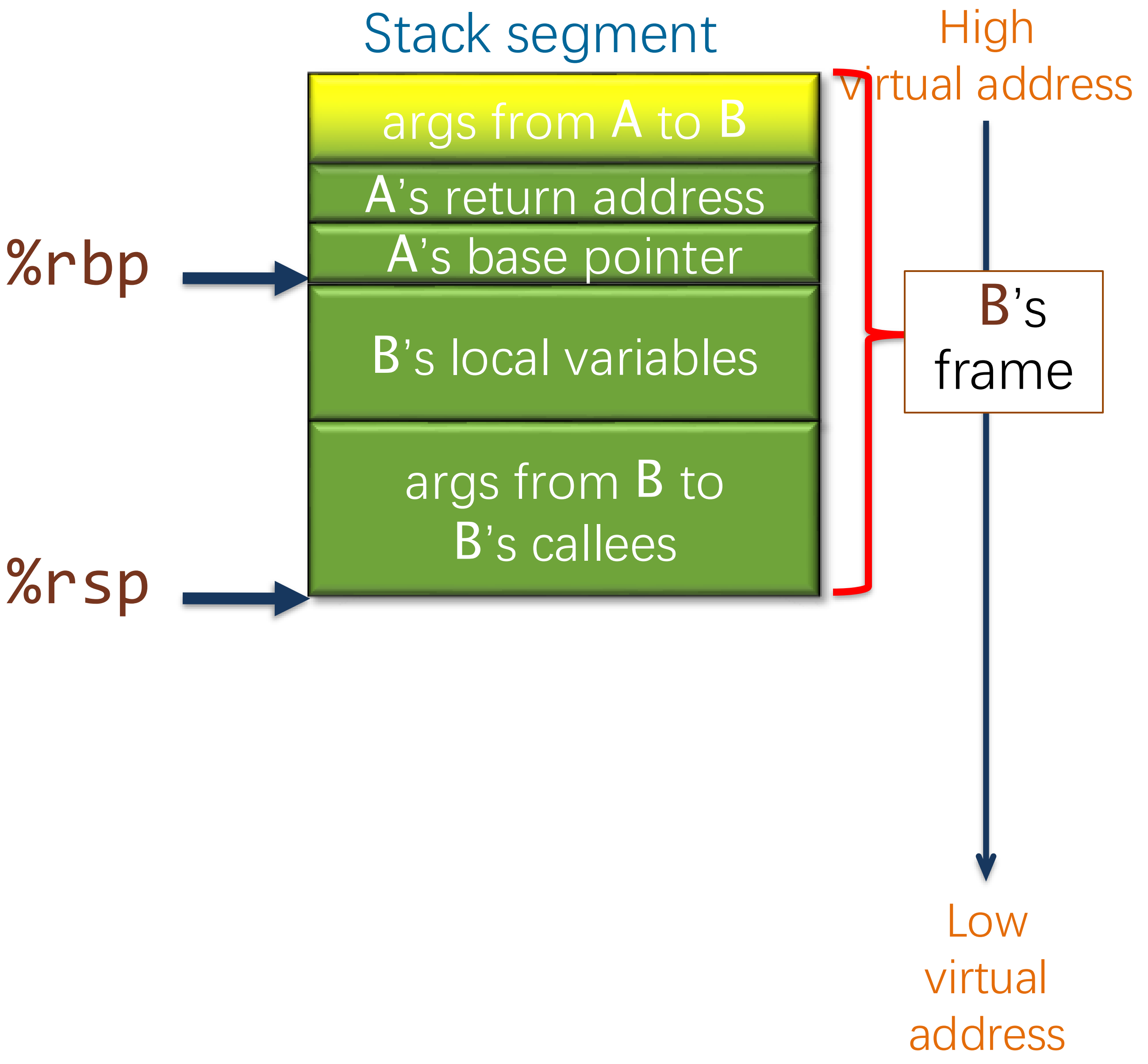
C linkage	64-bit name	32-bit **name**	16-bit name	8-bit name(s)
Return value	%rax	%eax	%ax	%ah, %al
Callee saved	%rbx	%ebx	%bx	%bh, %bl
4th argument	%rcx	%ecx	%cx	%ch, %cl
3rd argument	%rdx	%edx	%dx	%dh, %dl
2nd argument	%rsi	%esi	%si	%sil
1st argument	%rdi	%edi	%di	%dil
Base pointer	%rbp	%ebp	%bp	%bpl
Stack pointer	%rsp	%esp	%sp	%spl
5th argument	%r8	%r8d	%r8w	%r8b
6th argument	%r9	%r9d	%r9w	%r9b
Callee saved	%r10	%r10d	%r10w	%r10b
For linking	%r11	%r11d	%r11w	%r11b
Callee saved	%r12	%r12d	%r12w	%r12b
Callee saved	%r13			
Callee saved	%r14			
Callee saved	%r15			

The %xmm0-%xmm7 registers are used for floating-point arguments.

Example: Linux C Subroutine Linkage

Let's work through an example function call in action.

SETUP: Function B was called from function A and is about to call function C.

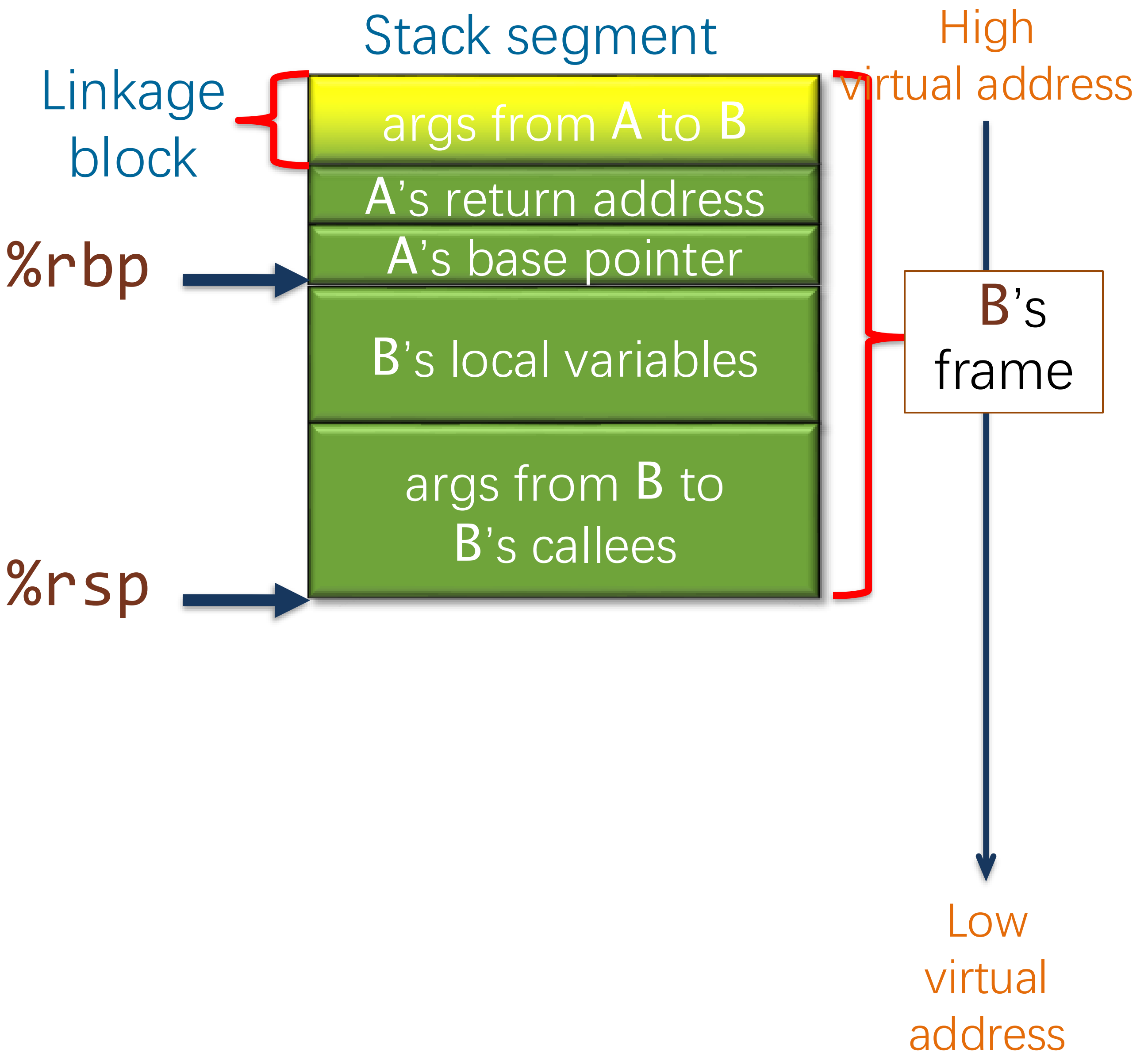


Example: Linux C Subroutine Linkage

Function B accesses its nonregister arguments from A,

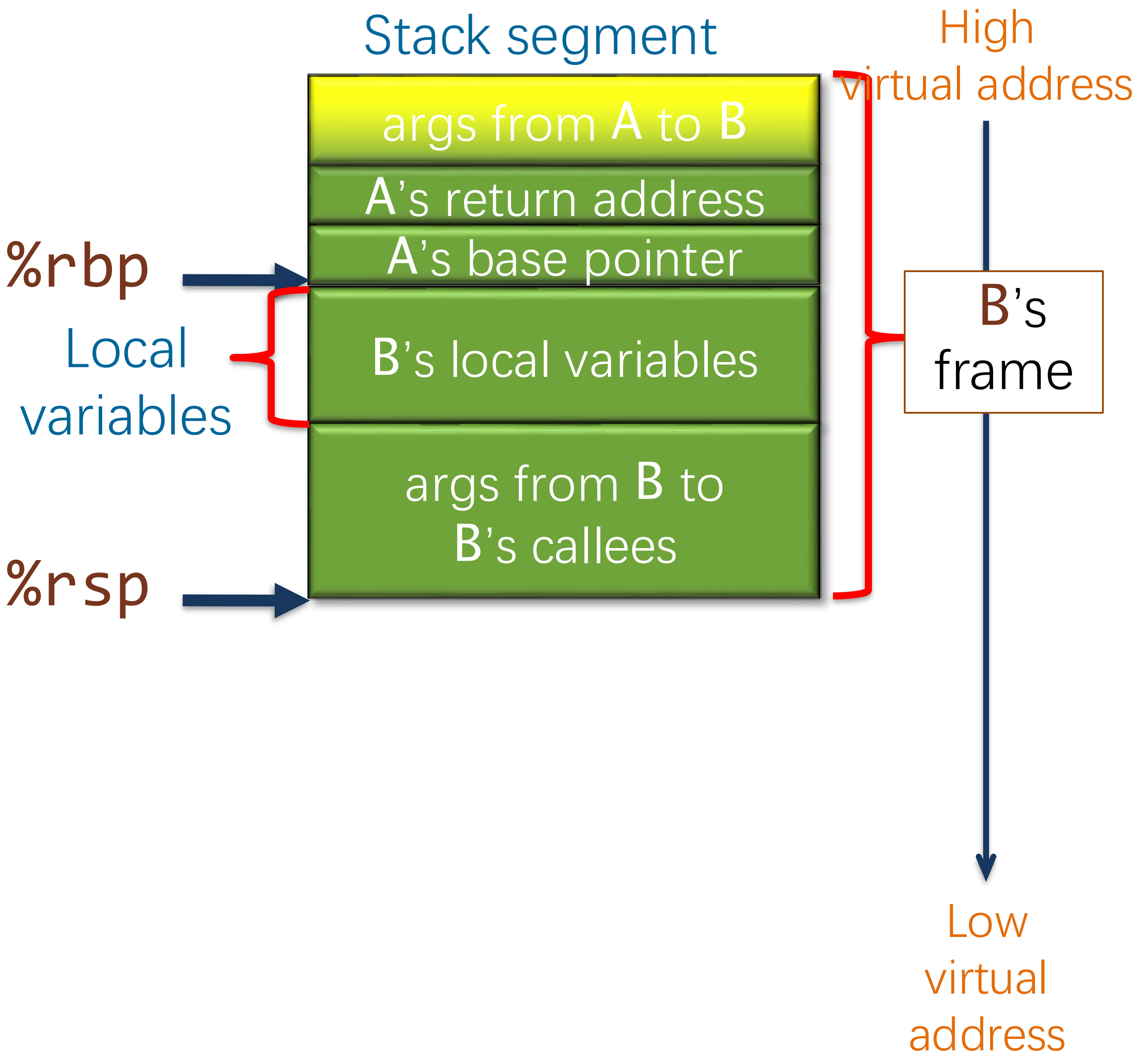
which lie in a linkage block,

by indexing `%rbp` with positive offsets.



Example: Linux C Subroutine Linkage

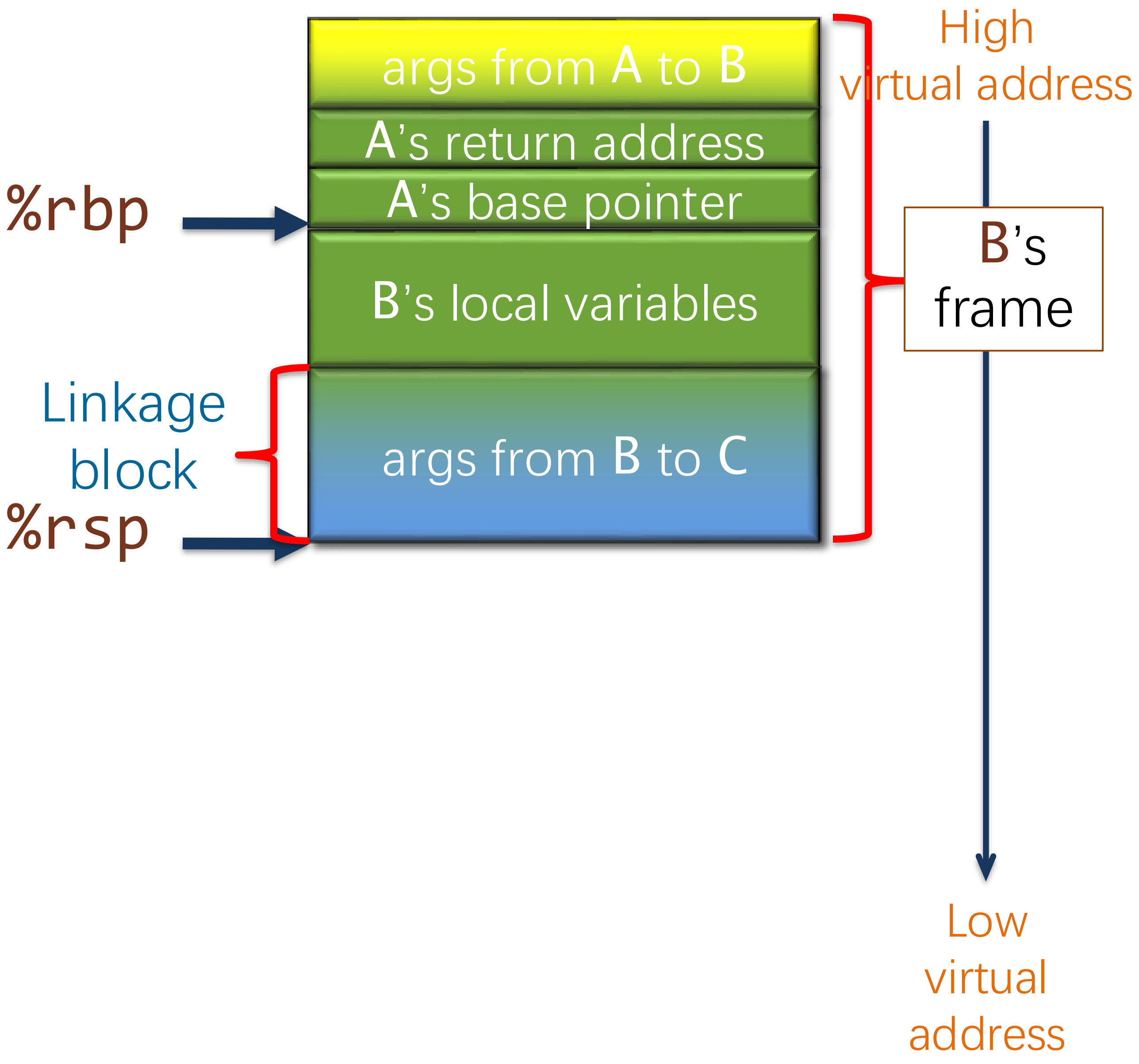
Function B accesses its local variables by indexing `%rbp` with negative offsets.



Example: Linux C Subroutine Linkage

Before B calls C,
B places the
nonregister arguments
for C into the reserved
linkage block it will
share with C,

which B accesses by
indexing %rbp with
negative offsets.

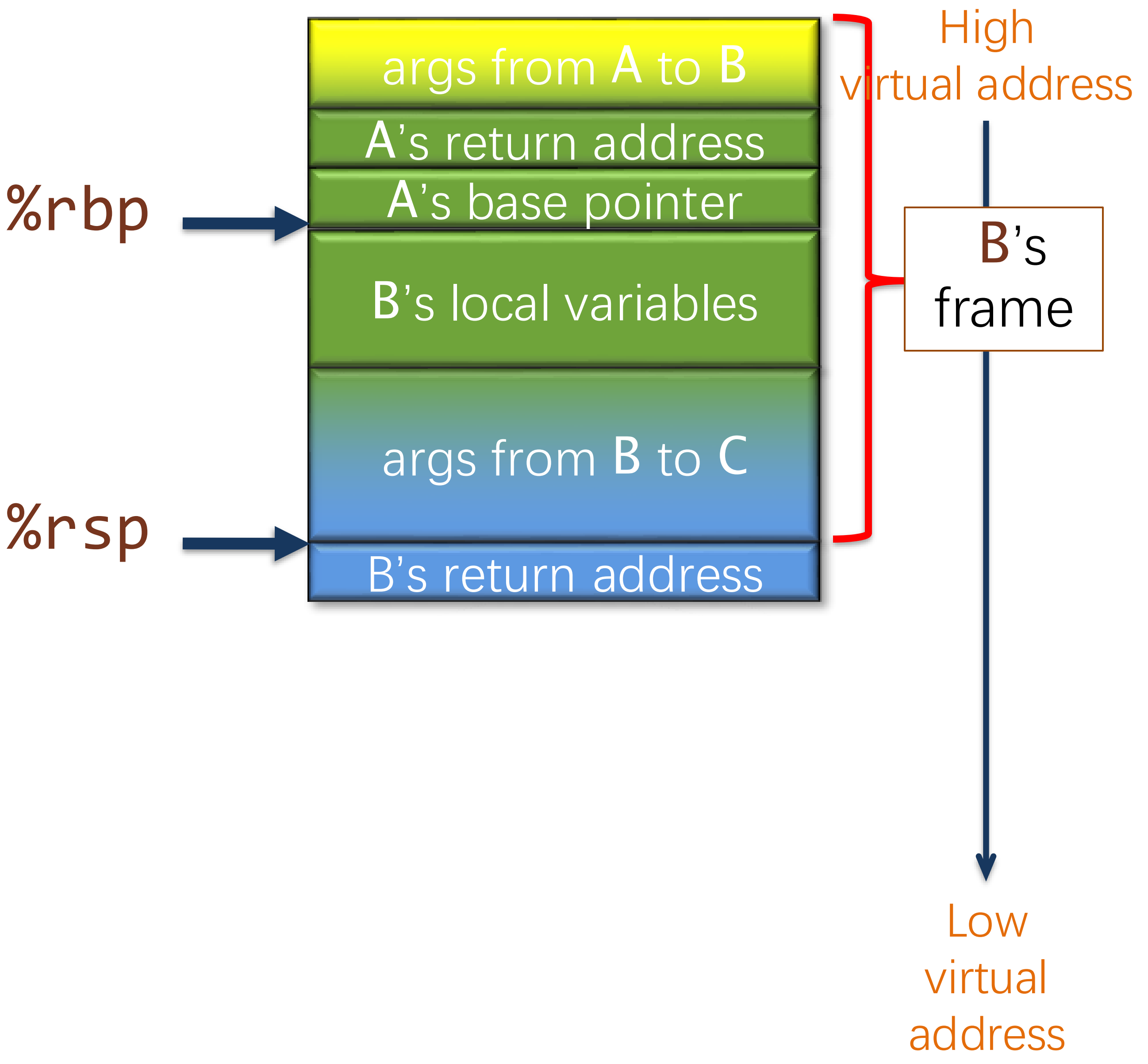


Example: Linux C Subroutine Linkage

Function B executes

```
call C
```

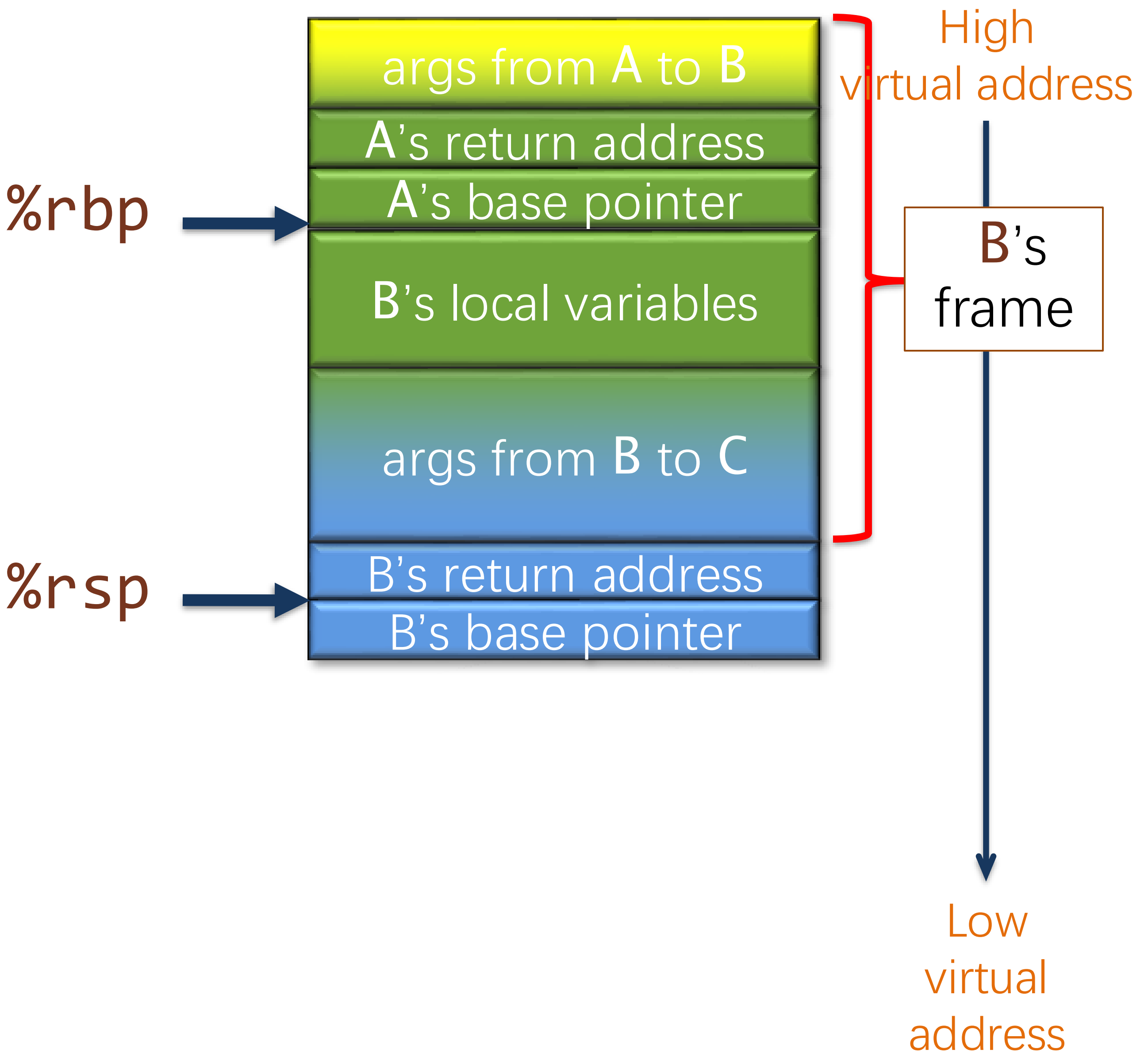
which pushes the return address for B on the stack and transfers control to C



Example: Linux C Subroutine Linkage

When function C starts, it executes a function prologue:

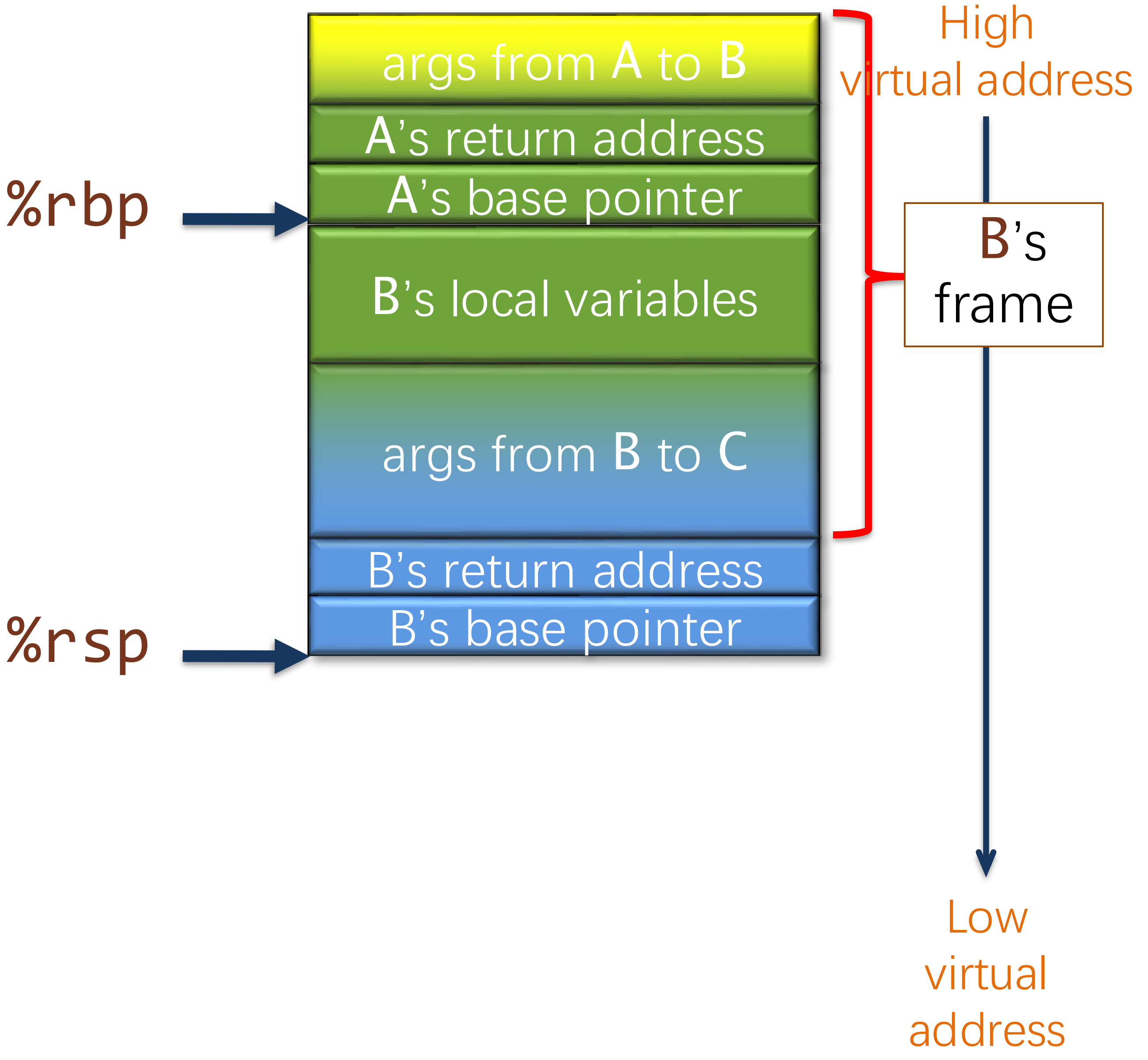
1. Push `%rbp` — B's base pointer — onto the stack,



Example: Linux C Subroutine Linkage

When function C starts, it executes a function prologue:

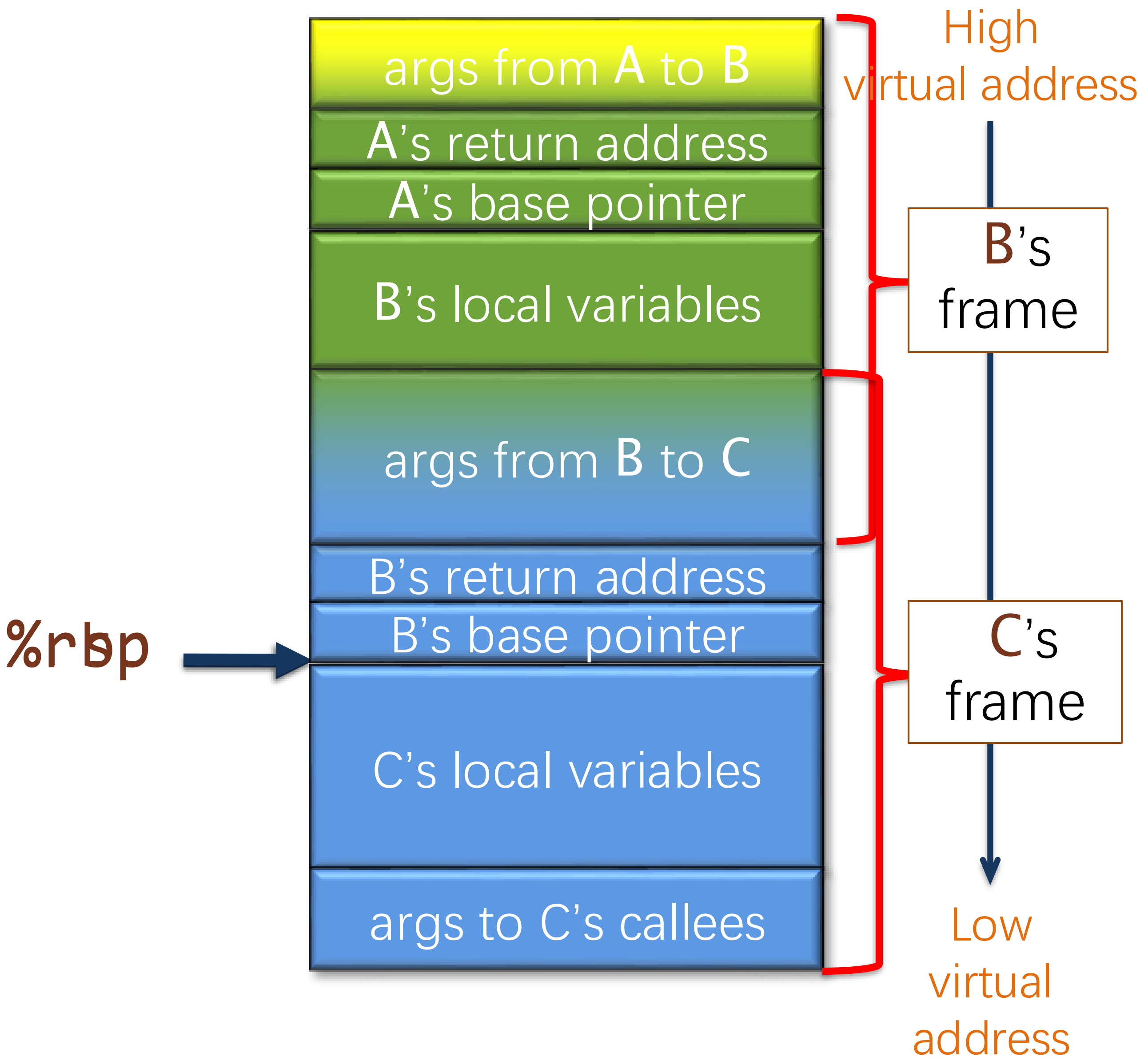
1. Push `%rbp` — B's base pointer — onto the stack,
2. Set `%rbp=%rsp`,



Example: Linux C Subroutine Linkage

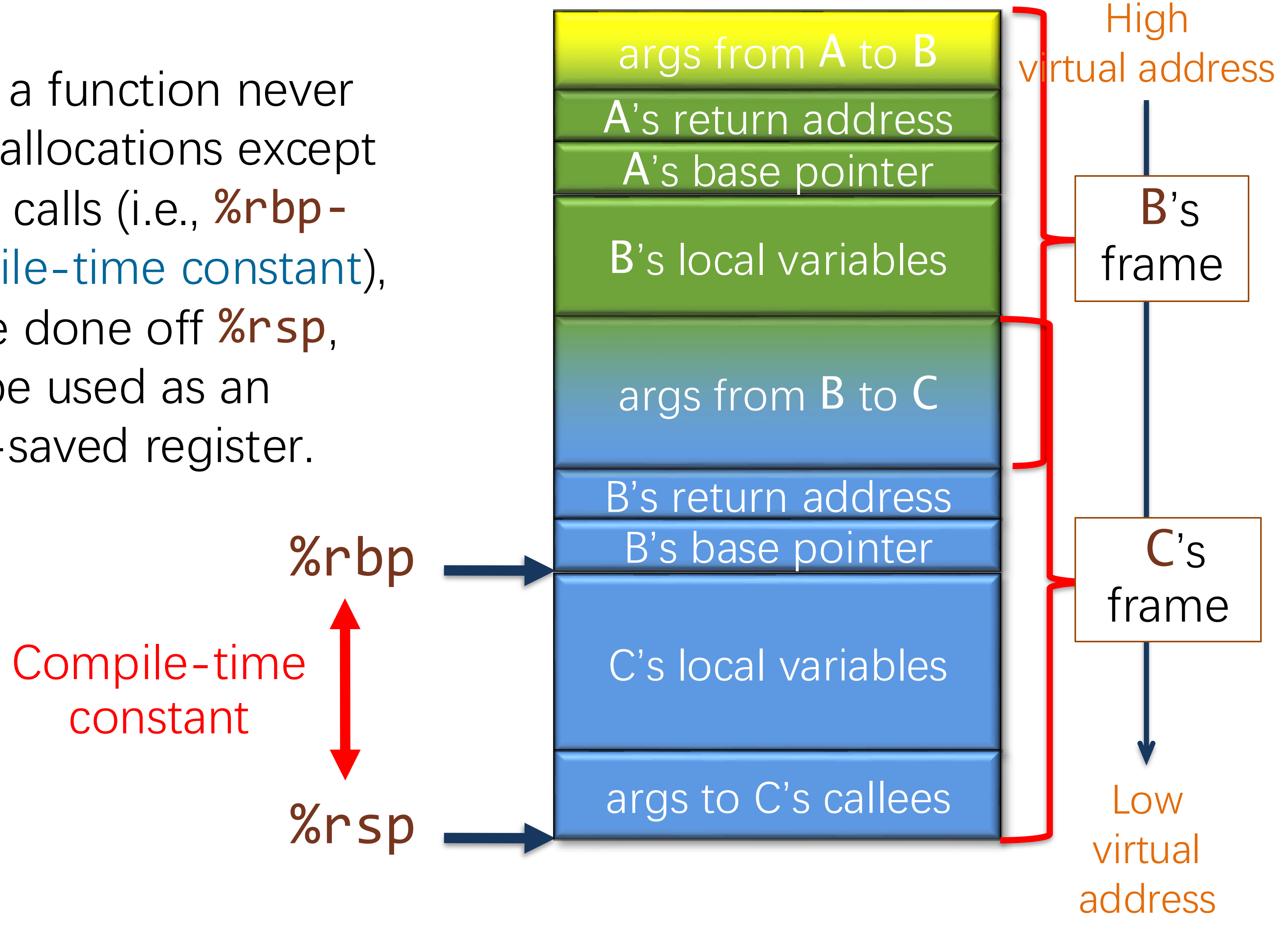
When function C starts, it executes a function prologue:

1. Push `%rbp` — B's base pointer — onto the stack,
2. Set `%rbp=%rsp`,
3. Advance `%rsp` to allocate space for C's local variables and linkage block.



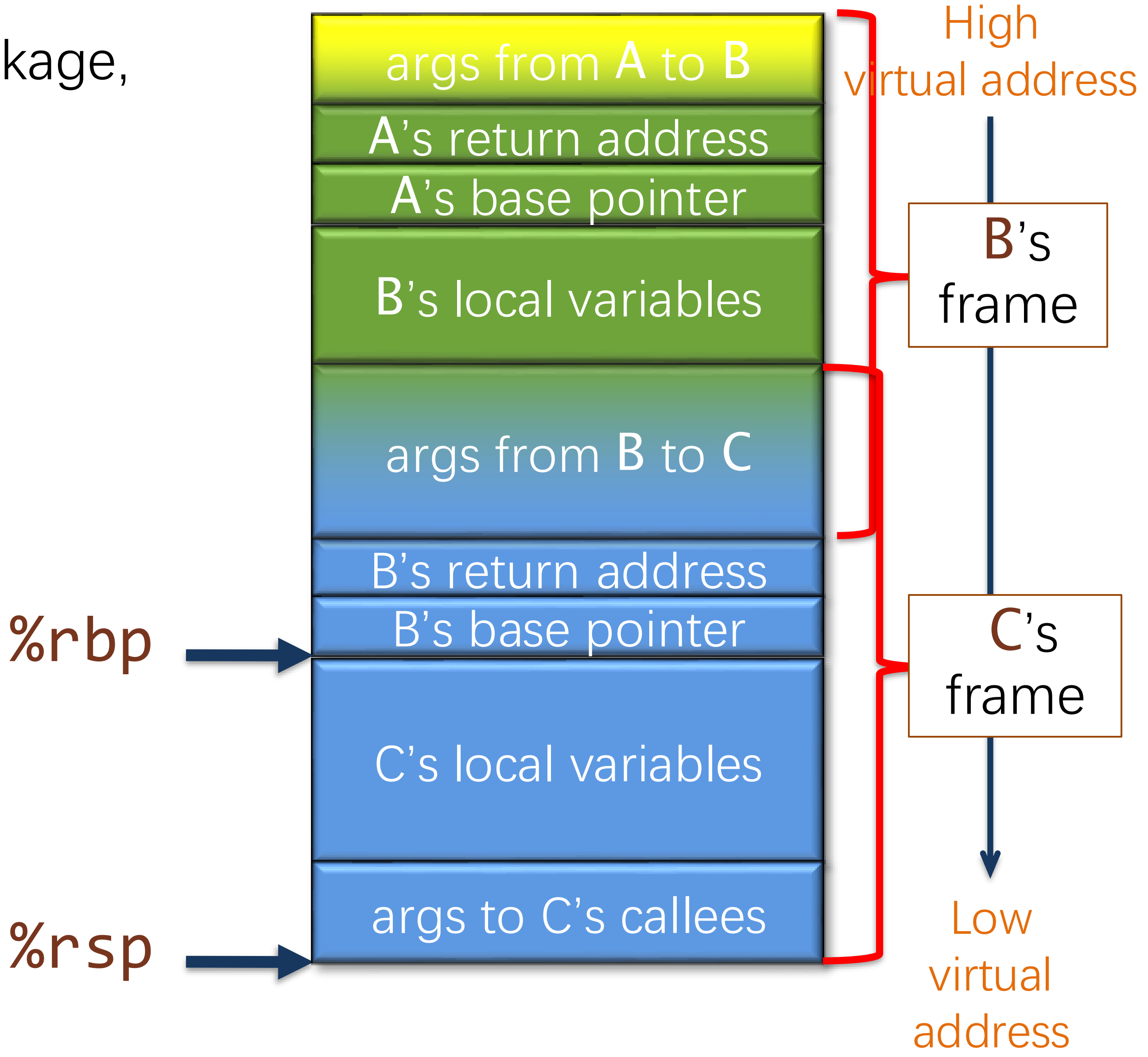
Example: Linux C Subroutine Linkage

OPTIMIZATION: If a function never performs stack allocations except during function calls (i.e., `%rbp-%rsp` is a compile-time constant), indexing can be done off `%rsp`, and `%rbp` can be used as an ordinary callee-saved register.



Example: Linux C Subroutine Linkage

For more details on C linkage, see the System V ABI.



PUTTING IT TOGETHER: FROM FIB.LL TO FIB.S

Compiling LLVM IR To Assembly

LLVM IR code `fib.ll`

```
define i64 @fib(i64 %0) {
  %2 = icmp slt i64 %0, 2
  br i1 %2, label %9, label %3

3:                          ; preds = %1
  %4 = add nsw i64 %0, -1
  %5 = call i64 @fib(i64 %4)
  %6 = add nsw i64 %0, -2
  %7 = call i64 @fib(i64 %6)
  %8 = add nsw i64 %7, %5
  br label %9

9:                          ; preds = %1, %3
  %10 = phi i64 [ %8, %3 ], [ %0, %1 ]
  ret i64 %10
}
```

Assembly code `fib.s`

```
        .globl      _fib
        .p2align    4, 0x90
_fib:   ## @fib
        pushq       %rbp
        movq        %rsp, %rbp
        pushq       %r14
        pushq       %rbx
        movq        %rdi, %rbx
        cmpq        $2, %rdi
        jl          LBB0_2
        leaq        -1(%rbx), %rdi
        callq       _fib
        movq        %rax, %r14
        addq        $-2, %rbx
        movq        %rbx, %rdi
        callq       _fib
        movq        %rax, %rbx
        addq        %r14, %rbx
LBB0_2:
        movq        %rbx, %rax
        popq        %rbx
        popq        %r14
        popq        %rbp
        retq
```

Roughly speaking, we can translate LLVM IR into assembly **line by line**.

From fib.ll to fib.s

LLVM IR code fib.ll

```
define i64 @fib(i64 %0) {
  %2 = icmp slt i64 %0, 2
  br i1 %2, label %9, label %3

3:                          ; preds = %1
  %4 = add nsw i64 %0, -1
  %5 = call i64 @fib(i64 %4)
  %6 = add nsw i64 %0, -2
  %7 = call i64 @fib(i64 %6)
  %8 = add nsw i64 %7, %5
  br label %9

9:                          ; preds = %1, %3
  %10 = phi i64 [ %8, %3 ], [ %0, %1 ]
  ret i64 %10
}
```

Assembly code fib.s

```
        .globl      _fib
        .p2align    4, 0x90
_fib:   ## @fib
        pushq       %rbp
        movq        %rsp, %rbp
        pushq       %r14
        pushq       %rbx
        movq        %rdi, %rbx
        cmpq        $2, %rdi
        jl          LBB0_2
        leaq        -1(%rbx), %rdi
        callq       _fib
        movq        %rax, %r14
        addq        $-2, %rbx
        movq        %rbx, %rdi
        callq       _fib
        movq        %rax, %rbx
        addq        %r14, %rbx
LBB0_2:
        movq        %rbx, %rax
        popq        %rbx
        popq        %r14
        popq        %rbp
        retq
```

```
        .globl      _fib
        .p2align    4, 0x90
_fib:           ## @fib
```

Declare the _fib label to be global

From fib.ll to fib.s

LLVM IR code fib.ll

```
define i64 @fib(i64 %0) {
  %2 = icmp slt i64 %0, 2
  br i1 %2, label %9, label %3

3:                          ; preds = %1
  %4 = add nsw i64 %0, -1
  %5 = call i64 @fib(i64 %4)
  %6 = add nsw i64 %0, -2
  %7 = call i64 @fib(i64 %6)
  %8 = add nsw i64 %7, %5
  br label %9

9:                          ; preds = %1, %3
  %10 = phi i64 [ %8, %3 ], [ %0, %1 ]
  ret i64 %10
}
```

Assembly code fib.s

```
        .globl      _fib
        .p2align    4, 0x90
_fib:   ## @fib
        pushq       %rbp
        movq        %rsp, %rbp
        pushq       %r14
        pushq       %rbx
        movq        %rdi, %rbx
        cmpq        $2, %rdi
        jl          LBB0_2
        leaq        -1(%rbx), %rdi
        callq       _fib
        movq        %rax, %r14
        addq        $-2, %rbx
        movq        %rbx, %rdi
        callq       _fib
        movq        %rax, %rbx
        addq        %r14, %rbx
LBB0_2:
        movq        %rbx, %rax
        popq        %rbx
        popq        %r14
        popq        %rbp
        retq
```

```
pushq   %rbp
movq    %rsp, %rbp
```

Function prologue: Save %rbp and sets %rbp = %rsp.

From `fib.ll` to `fib.s`

LLVM IR code `fib.ll`

```
define i64 @fib(i64 %0) {
  %2 = icmp slt i64 %0, 2
  br i1 %2, label %9, label %3

3:                          ; preds = %1
  %4 = add nsw i64 %0, -1
  %5 = call i64 @fib(i64 %4)
  %6 = add nsw i64 %0, -2
  %7 = call i64 @fib(i64 %6)
  %8 = add nsw i64 %7, %5
  br label %9

9:                          ; preds = %1, %3
  %10 = phi i64 [ %8, %3 ], [ %0, %1 ]
  ret i64 %10
}
```

Assembly code `fib.s`

```
        .globl      _fib
        .p2align    4, 0x90
_fib:   ## @fib
        pushq       %rbp
        movq        %rsp, %rbp
        pushq       %r14
        pushq       %rbx
        movq        %rdi, %rbx
        cmpq        $2, %rdi
        jl          LBB0_2
        leaq        -1(%rbx), %rdi
        callq       _fib
        movq        %rax, %r14
        addq        $-2, %rbx
        movq        %rbx, %rdi
        callq       _fib
        movq        %rax, %rbx
        addq        %r14, %rbx
LBB0_2:
        movq        %rbx, %rax
        popq        %rbx
        popq        %r14
        popq        %rbp
        retq
```

```
pushq   %r14
pushq   %rbx
```

Save any callee-saved registers that `fib` will use.

From fib.ll to fib.s

LLVM IR code fib.ll

```
define i64 @fib(i64 %0) {
  %2 = icmp slt i64 %0, 2
  br i1 %2, label %9, label %3

3:                          ; preds = %1
  %4 = add nsw i64 %0, -1
  %5 = call i64 @fib(i64 %4)
  %6 = add nsw i64 %0, -2
  %7 = call i64 @fib(i64 %6)
  %8 = add nsw i64 %7, %5
  br label %9

9:                          ; preds = %1, %3
  %10 = phi i64 [ %8, %3 ], [ %0, %1 ]
  ret i64 %10
}
```

```
movq    %rdi, %rbx
```

Assembly code fib.s

```
        .globl      _fib
        .p2align    4, 0x90
_fib:   ## @fib
        pushq       %rbp
        movq        %rsp, %rbp
        pushq       %r14
        pushq       %rbx
        movq        %rdi, %rbx
        cmpq        $2, %rdi
        jl          LBB0_2
        leaq        -1(%rbx), %rdi
        callq       _fib
        movq        %rax, %r14
        addq        $-2, %rbx
        movq        %rbx, %rdi
        callq       _fib
        movq        %rax, %rbx
        popq        %r14
        popq        %rbp
        retq
```

Register %rdi stores the function argument n.

Copy the incoming argument n into %rbx.

From `fib.ll` to `fib.s`

LLVM IR code `fib.ll`

```
define i64 @fib(i64 %0) {
  %2 = icmp slt i64 %0, 2
  br i1 %2, label %9, label %3

3:                          ; preds = %1
  %4 = add nsw i64 %0, -1
  %5 = call i64 @fib(i64 %4)
  %6 = add nsw i64 %0, -2
  %7 = call i64 @fib(i64 %6)
  %8 = add nsw i64 %7, %5
  br label %9

9:                          ; preds = %1, %3
  %10 = phi i64 [ %8, %3 ], [ %0, %1 ]
  ret i64 %10
}
```

```
cmpq    $2, %rbx
```

Assembly code `fib.s`

```
        .globl      _fib
        .p2align    4, 0x90
_fib:   ## @fib
        pushq       %rbp
        movq        %rsp, %rbp
        pushq       %r14
        pushq       %rbx
        movq        %rdi, %rbx
        cmpq        $2, %rdi
        jl          LBB0_2
        leaq        -1(%rbx), %rdi
        callq       _fib
        movq        %rax, %r14
        addq        $-2, %rbx
        movq        %rbx, %rdi
        callq       _fib
        movq        %rax, %rbx
        addq        %r14, %rbx
LBB0_2:
        movq        %rbx, %rax
        popq        %rbx
        popq        %r14
        popq        %rbp
        retq
```

Compare n against 2.

From fib.ll to fib.s

LLVM IR code fib.ll

```
define i64 @fib(i64 %0) {
  %2 = icmp slt i64 %0, 2
  br i1 %2, label %9, label %3

3:                          ; preds = %1
  %4 = add nsw i64 %0, -1
  %5 = call i64 @fib(i64 %4)
  %6 = add nsw i64 %0, -2
  %7 = call i64 @fib(i64 %6)
  %8 = add nsw i64 %7, %5
  br label %9

9:                          ; preds = %1, %3
  %10 = phi i64 [ %8, %3 ], [ %0, %1 ]
  ret i64 %10
}
```

```
jl      LBB0_2
```

Assembly code fib.s

```
        .globl      _fib
        .p2align    4, 0x90
_fib:   ## @fib
        pushq       %rbp
        movq        %rsp, %rbp
        pushq       %r14
        pushq       %rbx
        movq        %rdi, %rbx
        cmpq        $2, %rdi
        jl          LBB0_2
        leaq        -1(%rbx), %rdi
        callq       _fib
        movq        %rax, %r14
        addq        $-2, %rbx
        movq        %rbx, %rdi
        callq       _fib
        movq        %rax, %rbx
        addq        %r14, %rbx
LBB0_2:
        movq        %rbx, %rax
        popq        %rbx
        popq        %r14
        popq        %rbp
        retq
```

True side of LLVM branch:
if n < 2, jump to label LBB0_2

From fib.ll to fib.s

LLVM IR code fib.ll

```
define i64 @fib(i64 %0) {
  %2 = icmp slt i64 %0, 2
  br i1 %2, label %9, label %3

3:                          ; preds = %1
  %4 = add nsw i64 %0, -1
  %5 = call i64 @fib(i64 %4)
  %6 = add nsw i64 %0, -2
  %7 = call i64 @fib(i64 %6)
  %8 = add nsw i64 %7, %5
  br label %9

9:                          ; preds = %1, %3
  %10 = phi i64 [ %8, %3 ], [ %0, %1 ]
  ret i64 %10
}
```

```
leaq -1(%rbx), %rdi
```

Assembly code fib.s

```
        .globl      _fib
        .p2align    4, 0x90
_fib:   ## @fib
        pushq       %rbp
        movq        %rsp, %rbp
        pushq       %r14
        pushq       %rbx
        movq        %rdi, %rbx
        cmpq        $2, %rdi
        jl          LBB0_2
        leaq        -1(%rbx), %rdi
        callq       _fib
        movq        %rax, %r14
        addq        $-2, %rbx
        movq        %rbx, %rdi
        callq       _fib
        movq        %rax, %rbx
        popq        %r14
        popq        %rbp
        retq
```

Compute n-1.
Store the result in %rdi.

The lea opcode evaluates the given address stores the result in the destination register. It does not access memory.

From fib.ll to fib.s

LLVM IR code fib.ll

```
define i64 @fib(i64 %0) {
  %2 = icmp slt i64 %0, 2
  br i1 %2, label %9, label %3

3:                          ; preds = %1
  %4 = add nsw i64 %0, -1
  %5 = call i64 @fib(i64 %4)
  %6 = add nsw i64 %0, -2
  %7 = call i64 @fib(i64 %6)
  %8 = add nsw i64 %7, %5
  br label %9

9:                          ; preds = %1, %3
  %10 = phi i64 [ %8, %3 ], [ %0, %1 ]
  ret i64 %10
}
```

```
callq   _fib
```

Assembly code fib.s

```
        .globl      _fib
        .p2align    4, 0x90
_fib:   ## @fib
        pushq       %rbp
        movq        %rsp, %rbp
        pushq       %r14
        pushq       %rbx
        movq        %rdi, %rbx
        cmpq        $2, %rdi
        jl          LBB0_2
        leaq        -1(%rbx), %rdi
        callq       _fib
        movq        %rax, %r14
        addq        $-2, %rbx
        movq        %rbx, %rdi
        callq       _fib
        movq        %rax, %rbx
        addq        %r14, %rbx
LBB0_2:
        movq        %rbx, %rax
        popq        %rbx
        popq        %r14
        popq        %rbp
        retq
```

Call fib. The argument n-1 has already been stored in %rdi.

From fib.ll to fib.s

LLVM IR code fib.ll

```
define i64 @fib(i64 %0) {
  %2 = icmp slt i64 %0, 2
  br i1 %2, label %9, label %3

3:                                ; preds = %1
  %4 = add nsw i64 %0, -1
  %5 = call i64 @fib(i64 %4)
  %6 = add nsw i64 %0, -2
  %7 = call i64 @fib(i64 %6)
  %8 = add nsw i64 %7, %5
  br label %9

9:                                ; preds = %1, %3
  %10 = phi i64 [ %8, %3 ], [ %0, %1 ]
  ret i64 %10
}
```

Assembly code fib.s

```
        .globl     _fib
        .p2align   4, 0x90
_fib:   ## @fib
        pushq      %rbp
        movq       %rsp, %rbp
        ...
        leaq       -1(%rbx), %rdi
        callq      _fib
        movq       %rax, %r14
        addq       $-2, %rbx
        movq       %rbx, %rdi
        callq      _fib
        movq       %rax, %rbx
        addq       %r14, %rbx
LBB0_2:
        movq       %rbx, %rax
        popq       %rbx
        popq       %r14
        popq       %rbp
        retq
```

Register %rax stores the result of the last function call

```
movq    %rax, %r14
```

Save the result of calling fib into %r14

From fib.ll to fib.s

LLVM IR code fib.ll

```
define i64 @fib(i64 %0) {
  %2 = icmp slt i64 %0, 2
  br i1 %2, label %9, label %3

3:                        ; preds = %1
  %4 = add nsw i64 %0, -1
  %5 = call i64 @fib(i64 %4)
  %6 = add nsw i64 %0, -2
  %7 = call i64 @fib(i64 %6)
  %8 = add nsw i64 %7, %5
  br label %9

9:                        ; preds = %1, %3
  %10 = phi i64 [ %8, %3 ], [ %0, %1 ]
  ret i64 %10
}
```

Assembly code fib.s

```
        .globl      _fib
        .p2align    4, 0x90
_fib:   ## @fib
        pushq       %rbp
        movq        %rsp, %rbp
        pushq       %r14
        pushq       %rbx
        movq        %rdi, %rbx
        cmpq        $2, %rdi
        jl          LBB0_2
        leaq        -1(%rbx), %rdi
        callq       _fib
        movq        %rax, %r14
        addq        $-2, %rbx
        movq        %rbx, %rdi
        callq       _fib
        movq        %rax, %rbx
        addq        %r14, %rbx
LBB0_2:
        movq        %rbx, %rax
        popq        %rbx
        popq        %r14
        popq        %rbp
        retq
```

```
addq    $-2, %rbx
movq    %rbx, %rdi
```

Compute n-2, then move the result into %rdi.

From fib.ll to fib.s

LLVM IR code fib.ll

```
define i64 @fib(i64 %0) {
  %2 = icmp slt i64 %0, 2
  br i1 %2, label %9, label %3

3:                          ; preds = %1
  %4 = add nsw i64 %0, -1
  %5 = call i64 @fib(i64 %4)
  %6 = add nsw i64 %0, -2
  %7 = call i64 @fib(i64 %6)
  %8 = add nsw i64 %7, %5
  br label %9

9:                          ; preds = %1, %3
  %10 = phi i64 [ %8, %3 ], [ %0, %1 ]
  ret i64 %10
}
```

Assembly code fib.s

```
        .globl      _fib
        .p2align    4, 0x90
_fib:   ## @fib
        pushq       %rbp
        movq        %rsp, %rbp
        pushq       %r14
        pushq       %rbx
        movq        %rdi, %rbx
        cmpq        $2, %rdi
        jl          LBB0_2
        leaq        -1(%rbx), %rdi
        callq       _fib
        movq        %rax, %r14
        addq        $-2, %rbx
        movq        %rbx, %rdi
        callq       _fib
        movq        %rax, %rbx
        addq        %r14, %rbx
LBB0_2:
        movq        %rbx, %rax
        popq        %rbx
        popq        %r14
        popq        %rbp
        retq
```

```
callq   _fib
movq    %rax, %rbx
```

Call fib. The argument n-2 is in %rdi. Store the result in %rbx.

From fib.ll to fib.s

LLVM IR code fib.ll

```
define i64 @fib(i64 %0) {
  %2 = icmp slt i64 %0, 2
  br i1 %2, label %9, label %3

3:                          ; preds = %1
  %4 = add nsw i64 %0, -1
  %5 = call i64 @fib(i64 %4)
  %6 = add nsw i64 %0, -2
  %7 = call i64 @fib(i64 %6)
  %8 = add nsw i64 %7, %5
  br label %9

9:                          ; preds = %1, %3
  %10 = phi i64 [ %8, %3 ], [ %0, %1 ]
  ret i64 %10
}
```

```
addq   %r14, %rbx
```

Assembly code fib.s

```
        .globl      _fib
        .p2align    4, 0x90
_fib:   ## @fib
        pushq       %rbp
        movq        %rsp, %rbp
        pushq       %r14
        pushq       %rbx
        movq        %rdi, %rbx
        cmpq        $2, %rdi
        jl          LBB0_2
        leaq        -1(%rbx), %rdi
        callq       _fib
        movq        %rax, %r14
        addq        $-2, %rbx
        movq        %rbx, %rdi
        callq       _fib
        movq        %rax, %rbx
        addq        %r14, %rbx
LBB0_2:
        movq        %rbx, %rax
        popq        %rbx
        popq        %r14
        popq        %rbp
        retq
```

Add the results of fib(n-1) and fib(n-2). Save the sum in %rbx.

From fib.ll to fib.s

LLVM IR code fib.ll

```
define i64 @fib(i64 %0) {
  %2 = icmp slt i64 %0, 2
  br i1 %2, label %9, label %3

3:                               ; preds = %1
  %4 = add nsw i64 %0, -1
  %5 = call i64 @fib(i64 %4)
  %6 = add nsw i64 %0, -2
  %7 = call i64 @fib(i64 %6)
  %8 = add nsw i64 %7, %5
  br label %9

9:                               ; preds = %1, %3
  %10 = phi i64 [ %8, %3 ], [ %0, %1 ]
  ret i64 %10
}
```

LBB0_2:

Assembly code fib.s

```
            .globl      _fib
            .p2align    4, 0x90
_fib:       ## @fib
            pushq       %rbp
            movq        %rsp, %rbp
            pushq       %r14
            pushq       %rbx
            movq        %rdi, %rbx
            cmpq        $2, %rdi
            jl          LBB0_2
            leaq        -1(%rbx), %rdi
            callq       _fib
            movq        %rax, %r14
            addq        $-2, %rbx
            movq        %rbx, %rdi
            callq       _fib
            movq        %rax, %rbx
            addq        %r14, %rbx
LBB0_2:
            movq        %rbx, %rax
            popq        %rbx
            popq        %r14
            popq        %rbp
            retq
```

Label for the true side of the LLVM branch

From `fib.ll` to `fib.s`

LLVM IR code `fib.ll`

```
define i64 @fib(i64 %0) {
  %2 = icmp slt i64 %0, 2
  br i1 %2, label %9, label %3

3:                          ; preds =
  %4 = add nsw i64 %0, -1
  %5 = call i64 @fib(i64 %4)
  %6 = add nsw i64 %0, -2
  %7 = call i64 @fib(i64 %6)
  %8 = add nsw i64 %7, %5
  br label %9

9:                          ; preds = %1, %3
  %10 = phi i64 [ %8, %3 ], [ %0, %1 ]
  ret i64 %10
}
```

Assembly code `fib.s`

```
          .globl     _fib
          .p2align   4, 0x90
_fib:     ## @fib
          pushq      %rbp
          movq       %rsp, %rbp
          pushq      %r14
                     %rbx
                     %rdi, %rbx
                     $2, %rdi
                     LBB0_2
                     -1(%rbx), %rdi
          callq      _fib
          movq       %rax, %r14
          addq       $-2, %rbx
          movq       %rbx, %rdi
          callq      _fib
          movq       %rax, %rbx
          addq       %r14, %rbx
LBB0_2:
          movq       %rbx, %rax
          popq       %rbx
          popq       %r14
          popq       %rbp
          retq
```

Wait! What happened to this unconditional branch?

In the assembly, `LBB0_2` ends up on the fall-through path, so no explicit `jmp` instruction is required.

From `fib.ll` to `fib.s`

LLVM IR code `fib.ll`

```
define i64 @fib(i64 %0) {
  %2 = icmp slt i64 %0, 2
  br i1 %2, label %9, label %3

3:                           ; preds = %1
  %4 = add nsw i64 %0, -1
  %5 = call i64 @fib(i64 %4)
  %6 = add nsw i64 %0, -2
  %7 = call i64 @fib(i64 %6)
  %8 = add nsw i64 %7, %5
  br label %9

9:                           ; preds = %1, %3
  %10 = phi i64 [ %8, %3 ], [ %0, %1 ]
  ret i64 %10
}
```

```
movq  %rbx, %rax
```

Assembly code `fib.s`

```
        .globl     _fib
        .p2align   4, 0x90
_fib:   ## @fib
        pushq      %rbp
        movq       %rsp, %rbp
        pushq      %r14
        pushq      %rbx
        movq       %rdi, %rbx
        cmpq       $2, %rdi
        jl         LBB0_2
        leaq       -1(%rbx), %rdi
        callq      _fib
        movq       %rax, %r14
        addq       $-2, %rbx
        movq       %rbx, %rdi
        callq      _fib
        movq       %rax, %rbx
        addq       %r14, %rbx
LBB0_2:
        movq       %rbx, %rax
        popq       %rbx
        popq       %r14
        popq       %rbp
        retq
```

Move the result from `%rbx` into `%rax`.

From fib.ll to fib.s

LLVM IR code fib.ll

```
define i64 @fib(i64 %0) {
  %2 = icmp slt i64 %0, 2
  br i1 %2, label %9, label %3

3:                            ; preds = %1
  %4 = add nsw i64 %0, -1
  %5 = call i64 @fib(i64 %4)
  %6 = add nsw i64 %0, -2
  %7 = call i64 @fib(i64 %6)
  %8 = add nsw i64 %7, %5
  br label %9

9:                            ; preds = %1, %3
  %10 = phi i64 [ %8, %3 ], [ %0, %1 ]
  ret i64 %10
}
```

Assembly code fib.s

```
        .globl      _fib
        .p2align    4, 0x90
_fib:   ## @fib
        pushq       %rbp
        movq        %rsp, %rbp
        ...
        callq       _fib
        movq        %rax, %r14
        addq        $-2, %rbx
        movq        %rbx, %rdi
        callq       _fib
        movq        %rax, %rbx
        addq        %r14, %rbx
LBB0_2:
        movq        %rbx, %rax
        popq        %rbx
        popq        %r14
        popq        %rbp
        retq
```

Function epilogue: Restore the callee-saved registers that fib used.

```
popq    %rbx
popq    %r14
popq    %rbp
```

From fib.ll to fib.s

LLVM IR code fib.ll

```
define i64 @fib(i64 %0) {
  %2 = icmp slt i64 %0, 2
  br i1 %2, label %9, label %3

3:                          ; preds = %1
  %4 = add nsw i64 %0, -1
  %5 = call i64 @fib(i64 %4)
  %6 = add nsw i64 %0, -2
  %7 = call i64 @fib(i64 %6)
  %8 = add nsw i64 %7, %5
  br label %9

9:                          ; preds = %1, %3
  %10 = phi i64 [ %8, %3 ], [ %0, %1 ]
  ret i64 %10
}
```

```
retq
```

Assembly code fib.s

```
        .globl      _fib
        .p2align    4, 0x90
_fib:   ## @fib
        pushq       %rbp
        movq        %rsp, %rbp
        pushq       %r14
        pushq       %rbx
        movq        %rdi, %rbx
        cmpq        $2, %rdi
        jl          LBB0_2
        leaq        -1(%rbx), %rdi
        callq       _fib
        movq        %rax, %r14
        addq        $-2, %rbx
        movq        %rbx, %rdi
        callq       _fib
        movq        %rax, %rbx
        addq        %r14, %rbx
LBB0_2:
        movq        %rbx, %rax
        popq        %rbx
        popq        %r14
        popq        %rbp
        retq
```

Return from the function

Summary: LLVM IR to Assembly

- To transform LLVM IR into assembly:
  - LLVM IR is translated approximately line by line into assembly
  - ISA instructions are chosen to implement primitive LLVM operations
  - ISA registers (and stack space) are allocated to store LLVM IR registers
  - Functions and function calls are implemented according to a calling convention

Summary: From C to Assembly

- We can reason through the mapping from C code to assembly in two steps: C to LLVM IR, and then LLVM IR to assembly

- LLVM IR organizes a C function into a control-flow graph
  - Nodes are basic blocks, which correspond with straight-line code in C
  - C control-flow constructs induce control-flow edges

- Assembly implements the LLVM IR code using ISA registers and the stack, according to a calling convention

References

- Quick reference on assembly instructions:
  - http://en.wikipedia.org/wiki/X86_instruction_listings
- Full details: Intel Software Developer Manual (on course site)
- C subroutine linkage: system V Application Binary Interface (on course site)
- LLVM IR language reference https://llvm.org/docs/LangRef.html
- Compiler intrinsic for inline assembly:
  - http://www.ibiblio.org/gferg/ldp/GCC-Inline-Assembly-HOWTO.html

Optional Slides

Putting It Together: From fib.c to fib.ll

From fib.c to fib.ll

C code fib.c

```c
int64_t fib(int64_t n)
{
  if (n < 2)
    return n;
  return
    fib(n-1)
    +
    fib(n-2);
}
```

LLVM IR fib.ll

```llvm
define i64 @fib(i64 %0) {
  %2 = icmp slt i64 %0, 2
  br i1 %2, label %9, label %3

3:
  %4 = add nsw i64 %0, -1
  %5 = call i64 @fib(i64 %4)
  %6 = add nsw i64 %0, -2
  %7 = call i64 @fib(i64 %6)
  %8 = add nsw i64 %7, %5
  br label %9

9:
  %10 = phi i64 [ %8, %3 ], [ %0, %1 ]
  ret i64 %10
}
```

From fib.c to fib.ll

C code fib.c

```c
int64_t fib(int64_t n)
{
  if (n < 2)
    return n;
  return
    fib(n-1)
    +
    fib(n-2);
}
```

LLVM IR fib.ll

```llvm
define i64 @fib(i64 %0) {
  %2 = icmp slt i64 %0, 2
  br i1 %2, label %9, label %3

3:
  %4 = add nsw i64 %0, -1
  %5 = call i64 @fib(i64 %4)
  %6 = add nsw i64 %0, -2
  %7 = call i64 @fib(i64 %6)
  %8 = add nsw i64 %7, %5
  br label %9

9:
  %10 = phi i64 [ %8, %3 ], [ %0, %1 ]
  ret i64 %10
}
```

From fib.c to fib.ll

C code fib.c

```c
int64_t fib(int64_t n)
{
  if (n < 2)
    return n;
  return
    fib(n-1)
    +
    fib(n-2);
}
```

LLVM IR fib.ll

```llvm
define i64 @fib(i64 %0) {
  %2 = icmp slt i64 %0, 2
  br i1 %2, label %9, label %3

3:
  %4 = add nsw i64 %0, -1
  %5 = call i64 @fib(i64 %4)
  %6 = add nsw i64 %0, -2
  %7 = call i64 @fib(i64 %6)
  %8 = add nsw i64 %7, %5
  br label %9

9:
  %10 = phi i64 [ %8, %3 ], [ %0, %1 ]
  ret i64 %10
}
```

From fib.c to fib.ll

C code fib.c

```c
int64_t fib(int64_t n)
{
  if (n < 2)
    return n;
  return
    fib(n-1)
    +
    fib(n-2);
}
```

LLVM IR fib.ll

```llvm
define i64 @fib(i64 %0) {
  %2 = icmp slt i64 %0, 2
  br i1 %2, label %9, label %3

3:
  %4 = add nsw i64 %0, -1
  %5 = call i64 @fib(i64 %4)
  %6 = add nsw i64 %0, -2
  %7 = call i64 @fib(i64 %6)
  %8 = add nsw i64 %7, %5
  br label %9

9:
  %10 = phi i64 [ %8, %3 ], [ %0, %1 ]
  ret i64 %10
}
```

From `fib.c` to `fib.ll`

C code `fib.c`

```c
int64_t fib(int64_t n)
{
  if (n < 2)
    return n;
  return
    fib(n-1)
    +
    fib(n-2);
}
```

LLVM IR `fib.ll`

```llvm
define i64 @fib(i64 %0) {
  %2 = icmp slt i64 %0, 2
  br i1 %2, label %9, label %3

3:
  %4 = add nsw i64 %0, -1
  %5 = call i64 @fib(i64 %4)
  %6 = add nsw i64 %0, -2
  %7 = call i64 @fib(i64 %6)
  %8 = add nsw i64 %7, %5
  br label %9

9:
  %10 = phi i64 [ %8, %3 ], [ %0, %1 ]
  ret i64 %10
}
```

From fib.c to fib.ll

C code fib.c

```c
int64_t fib(int64_t n)
{
  if (n < 2)
    return n;
  return
    fib(n-1)
    +
    fib(n-2);
}
```

LLVM IR fib.ll

```llvm
define i64 @fib(i64 %0) {
  %2 = icmp slt i64 %0, 2
  br i1 %2, label %9, label %3

3:
  %4 = add nsw i64 %0, -1
  %5 = call i64 @fib(i64 %4)
  %6 = add nsw i64 %0, -2
  %7 = call i64 @fib(i64 %6)
  %8 = add nsw i64 %7, %5
  br label %9

9:
  %10 = phi i64 [ %8, %3 ], [ %0, %1 ]
  ret i64 %10
}
```

From fib.c to fib.ll

C code fib.c

```c
int64_t fib(int64_t n)
{
  if (n < 2)
    return n;
  return
    fib(n-1)
    +
    fib(n-2);
}
```

LLVM IR fib.ll

```llvm
define i64 @fib(i64 %0) {
  %2 = icmp slt i64 %0, 2
  br i1 %2, label %9, label %3

3:
  %4 = add nsw i64 %0, -1
  %5 = call i64 @fib(i64 %4)
  %6 = add nsw i64 %0, -2
  %7 = call i64 @fib(i64 %6)
  %8 = add nsw i64 %7, %5
  br label %9

9:
  %10 = phi i64 [ %8, %3 ], [ %0, %1 ]
  ret i64 %10
}
```

From fib.c to fib.ll

C code fib.c

```c
int64_t fib(int64_t n)
{
  if (n < 2)
    return n;
  return
    fib(n-1)
    +
    fib(n-2);
}
```

LLVM IR fib.ll

```llvm
define i64 @fib(i64 %0) {
  %2 = icmp slt i64 %0, 2
  br i1 %2, label %9, label %3

3:
  %4 = add nsw i64 %0, -1
  %5 = call i64 @fib(i64 %4)
  %6 = add nsw i64 %0, -2
  %7 = call i64 @fib(i64 %6)
  %8 = add nsw i64 %7, %5
  br label %9

9:
  %10 = phi i64 [ %8, %3 ], [ %0, %1 ]
  ret i64 %10
}
```

Handling LLVM IR Manually

Viewing LLVM IR Manually

You can see what the `clang` compiler does by looking at the LLVM IR.

Source code `fib.c`

```c
int64_t fib(int64_t n) {
  if (n < 2) return n;
  return fib(n-1) + fib(n-2);
}
```

```
$ clang -O3 fib.c \
> -S -emit-llvm
```

Clang flags:

- "`-S`" produces assembly.
- "`-S -emit-llvm`" produces LLVM IR.

LLVM IR code `fib.ll`

```llvm
define i64 @fib(i64 %0) local_unnamed_addr #0 {
  %2 = icmp slt i64 %0, 2
  br i1 %2, label %9, label %3

3:                                    ; preds = %1
  %4 = add nsw i64 %0, -1
  %5 = call i64 @fib(i64 %4)
  %6 = add nsw i64 %0, -2
  %7 = call i64 @fib(i64 %6)
  %8 = add nsw i64 %7, %5
  br label %9

9:                                    ; preds = %1, %3
  %10 = phi i64 [ %8, %3 ], [ %0, %1 ]
  ret i64 %10
}
```

Compiling LLVM IR Manually

LLVM IR can be translated directly into assembly.

LLVM IR code fib.ll

```
define i64 @fib(i64 %0) local_unnamed_addr #0 {
  %2 = icmp slt i64 %0, 2
  br i1 %2, label %9, label %3

3:                                  ; preds = %1
  %4 = add nsw i64 %0, -1
  %5 = call i64 @fib(i64 %4)
  %6 = add nsw i64 %0, -2
  %7 = call i64 @fib(i64 %6)
  %8 = add nsw i64 %7, %5
  br label %9

9:                                  ; preds = %1, %3
  %10 = phi i64 [ %8, %3 ], [ %0, %1 ]
  ret i64 %10
}
```

```
$ clang fib.ll -S
```

Assembly code fib.s

```
            .globl      _fib
            .p2align    4, 0x90
_fib:       ## @fib
            pushq       %rbp
            movq        %rsp, %rbp
            pushq       %r14
            pushq       %rbx
            movq        %rdi, %rbx
            cmpq        $2, %rdi
            jl          LBB0_2
            leaq        -1(%rbx), %rdi
            callq       _fib
            movq        %rax, %r14
            addq        $-2, %rbx
            movq        %rbx, %rdi
            callq       _fib
            movq        %rax, %rbx
            addq        %r14, %rbx
LBB0_2:
            movq        %rbx, %rax
            popq        %rbx
            popq        %r14
            popq        %rbp
            retq
```

LLVM IR Overview

LLVM IR Registers

LLVM IR stores values in registers.

- Syntax: %<name>
- LLVM registers are like local variables in C: LLVM supports an infinite number of registers, each distinguished by name.
- Register names are local to each LLVM IR function.

Registers in an LLVM IR snippet.

```
%4 = add nsw i64 %0, -1
%5 = call i64 @fib(i64 %4)
%6 = add nsw i64 %0, -2
%7 = call i64 @fib(i64 %6)
%8 = add nsw i64 %7, %5
br label %9
```

One catch: We shall see that LLVM hijacks its syntax for registers to refer to "basic blocks."

LLVM IR Instructions

LLVM-IR code is organized into instructions.

- Syntax for instructions that produce a value:
```
%<name> = <opcode> [flags] <operands>
```
- Syntax for other instructions:
```
<opcode> <operands>
```
- Operands are registers, constants, or “basic blocks,” often with their data type.

Instruction that produces a value.

Instruction that does not produce a value.

```
%4 = add nsw i64 %0, -1
%5 = call i64 @fib(i64 %4)
%6 = add nsw i64 %0, -2
%7 = call i64 @fib(i64 %6)
%8 = add nsw i64 %7, %5
br label %9
```

Common LLVM IR Instructions

Type or operation		Example(s)
Data movement	Stack allocation	alloca
	Memory read	load
	Memory write	store
	Type conversion	bitcast, ptrtoint
Arithmetic and logic	Integer arithmetic	add, sub, mul, div, shl, shr
	Floating-point arithmetic	fadd, fmul
	Binary logic	and, or, xor, not
	Boolean logic	icmp
	Address calculation	getelementptr
Control flow	Unconditional branch	br <location>
	Conditional branch	br <condition>, <true>, <false>
	Subroutines	call, ret
	Maintaining SSA form	phi

LLVM IR Data Types

LLVM IR uses a simple type system.

- Integers: i<number>
  - Example: A 64-bit integer: i64
  - Example: A 1-bit integer: i1
- Floating-point values: double, float
- Arrays: [<number> x <type>]
  - Example: An array of 5 int’s: [5 x i32]
- Structs: { <type>, … }
- Vectors: < <number> x <type> >
- Pointers: <type>*
  - Example: A pointer to an 8-bit integer: i8*
- Labels (i.e., basic blocks): label

C Loops to LLVM IR

Components of a C Loop

```c
void dax(
     double *restrict y, double a,
     const double *restrict x,
     int64_t n) {
   for (int64_t i = 0; i < n; ++i)
     y[i] = a * x[i];
}
```

C code

A C loop involves a loop body and loop control.

Loop body

Loop control

LLVM IR snippet

```llvm
8:                                  ; preds = %6, %8
  %9 = phi i64 [ %14, %8 ], [ 0, %6 ]
  %10 = getelementptr inbounds double,
        double* %2, i64 %9
  %11 = load double, double* %10, align 8
  %12 = fmul double %11, %1
  %13 = getelementptr inbounds double,
        double* %0, i64 %9
  store double %12, double* %13, align 8
  %14 = add nuw nsw i64 %9, 1
  %15 = icmp eq i64 %14, %3
  br i1 %15, label %7, label %8
```

Loops in the CFG

A C loop produces a loop pattern in the control-flow graph.

C code

```
for (int64_t i = 0;
       i < n;
       ++i)
  …
return;
```

Early test of
n > 0.

Control-flow graph

```
%5 = icmp sgt i64 %3, 0
br i1 %5, label %8, label %7
```

```
8:                         ; preds = %6, %8
  %9 = phi i64 [ %14, %8 ], [ 0, %6 ]
  …
  %14 = add nuw nsw i64 %9, 1
  %15 = icmp eq i64 %14, %3
  br i1 %15, label %7, label %8
```

Loop block has 2
incoming edges.

Exit from
the loop.

```
7:                         ; preds = %8, %6
  ret void
```

Back edge

Loop Control

The loop control for a C loop consists of a loop induction variable, an initialization, a condition, and an increment.

Register %3 holds the value of n.

C code

```
for (int64_t i = 0; i < n; ++i)
  …
```

Initialization

Increment

Condition

LLVM IR

```
8:                          ; preds = %6, %8
  %9 = phi i64 [ %14, %8 ], [ 0, %6 ]
  …
  %14 = add nuw nsw i64 %9, 1
  %15 = icmp eq i64 %14, %3
  br i1 %15, label %7, label %8
```

Because of SSA, the induction variable occupies multiple registers.

Loop Induction Variables

The induction variable changes registers at the code for the loop increment.

```c
for (int64_t i = 0; i < n; ++i)
  y[i] = a * x[i];
```

C code

LLVM IR

```llvm
8:                                          ; preds = %6, %8
  %9 = phi i64 [ %14, %8 ], [ 0, %6 ]
  %10 = getelementptr inbounds double,
        double* %2, i64 %9
  %11 = load double, double* %10, align 8
  %12 = fmul double %11, %1
  %13 = getelementptr inbounds double,
        double* %0, i64 %9
  store double %12, double* %13, align 8
  %14 = add nuw nsw i64 %9, 1
  %15 = icmp eq i64 %14, %3
  br i1 %15, label %7, label %8
```

The induction variable gets either the initial or incremented value.

The incremented value ends up in a new register.

MEMORY OPERATIONS

Dealing with Memory

Operations on pointers and aggregate types — arrays or structs — generally involve accessing memory.

A memory access in LLVM IR typically involves computing an address followed by reading or writing memory.

C code

```c
int* A;
A[i];
```

LLVM IR

```llvm
%3 = getelementptr inbounds i32,
     i32* %0, i64 %1
%4 = load i32, i32* %3, align 4
```

Compute an address and store it into register %5.

Read memory at the address stored in %5.

The getelementptr Instruction

The getelementptr instruction computes a memory address from a pointer and a list of indices.

C code

```
int* A;
A[i];
```

Example: Compute the address `i32* %0 + %1` using pointer arithmetic.

```
%3 = getelementptr inbounds i32,
     i32* %0, i64 %1
```

Pointer to the memory for A

Index i

See `https://llvm.org/docs/GetElementPtr.html`

LLVM IR ATTRIBUTES

Attributes

LLVM IR constructs — including instructions, operands, functions, and function parameters — might be decorated with attributes.

C code

```c
const uint64_t deBruijn = 0x022fdd63cc95386d;
const int convert[64] = { … };
int r = convert[(x * deBruijn) >> 58];
```

LLVM IR

```llvm
%4 = getelementptr inbounds [64 x i32],
[64 x i32]* @convert, i64 0, i64 %3
%5 = load i32, i32* %4, align 4, !tbaa !2
```

Attribute describing the alignment of the read from memory.

Where Do Attributes Come From?

Some attributes are derived from the source code.

C code daxpy.c

```
void daxpy(…
           const double *restrict x,
           …)
```

LLVM IR daxpy.ll

```
define void @daxpy(
  …
  double* noalias nocapture readonly,
  …)
```

Other attributes are determined by compiler analysis.

Analysis determined the alignment of this read.

LLVM IR

```
%15 = load double, double* %14, align 8
```

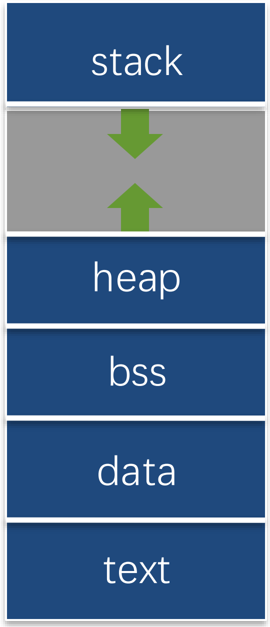
Summary of LLVM IR

LLVM IR is similar to assembly, but simpler

Assembler Directives

Assembler Directives

Assembly code contains directives that refer to and operate on sections of assembly.



- Segment directives organize the contents of an assembly file into segments.
  - "`.text`": Identifies the text segment.
  - "`.bss`": Identifies the bss segment.
  - "`.data`": Identifies the data segment.
- Storage directives store content into the current segment.

  Examples:

`x: .space 20`	Allocates 20 bytes at location x.
`y: .long 172`	Stores the constant 172L at location y.
`z: .asciz "6.172"`	Stores the string "6.172\0" at location z.
`.align 8`	Align the next content to an 8-byte boundary.

- Scope and linkage directives control linking.

  Example: "`.globl fib`": Makes "`fib`" visible to other object files.